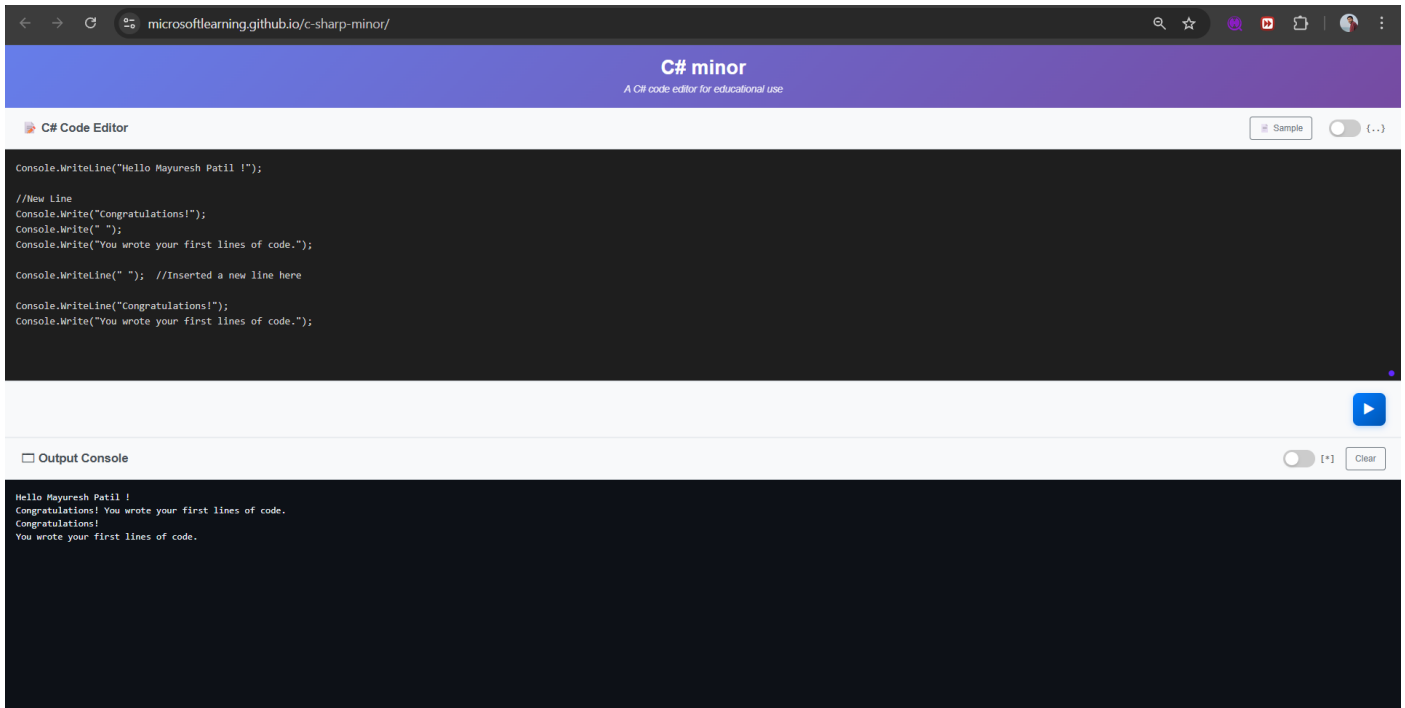


Develop Apps with .NET

First C# code :



The screenshot shows the C# minor code editor interface. The browser address bar displays `microsoftlearning.github.io/c-sharp-minor/`. The editor title is "C# minor" with the subtitle "A C# code editor for educational use". The code editor area contains the following C# code:

```
Console.WriteLine("Hello Mayuresh Patil !");

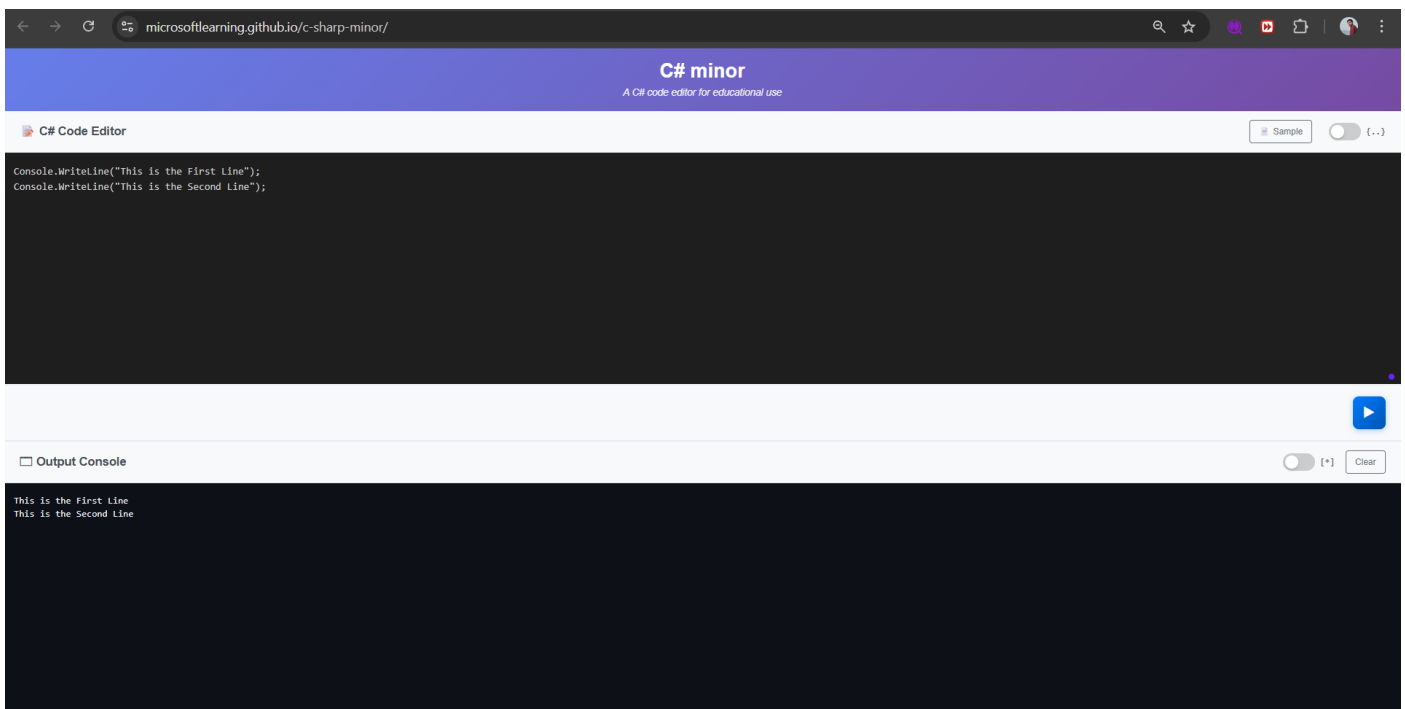
//New Line
Console.WriteLine("Congratulations!");
Console.Write(" ");
Console.WriteLine("You wrote your first lines of code.");

Console.WriteLine(" "); //Inserted a new line here

Console.WriteLine("Congratulations!");
Console.WriteLine("You wrote your first lines of code.");
```

Below the code editor is the "Output Console" panel, which shows the execution results:

```
Hello Mayuresh Patil !
Congratulations! You wrote your first lines of code.
Congratulations!
You wrote your first lines of code.
```



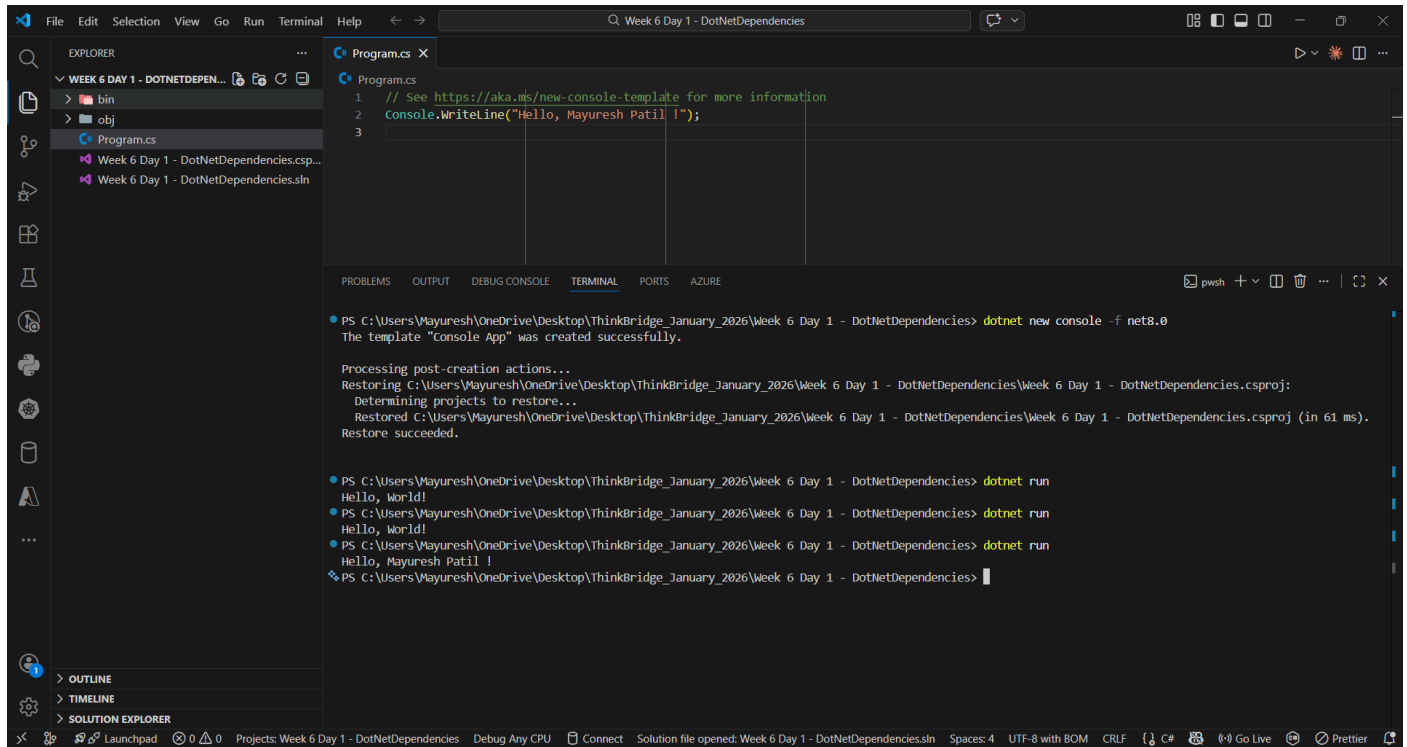
The screenshot shows the C# minor code editor interface. The browser address bar displays `microsoftlearning.github.io/c-sharp-minor/`. The editor title is "C# minor" with the subtitle "A C# code editor for educational use". The code editor area contains the following C# code:

```
Console.WriteLine("This is the First Line");
Console.WriteLine("This is the Second Line");
```

Below the code editor is the "Output Console" panel, which shows the execution results:

```
This is the First Line
This is the Second Line
```

Exercise - Install packages



The screenshot shows the Visual Studio Code interface with a project named "Week 6 Day 1 - DotNetDependencies". The Explorer pane on the left shows the project structure with a "bin" folder, an "obj" folder, and a "Program.cs" file. The Program.cs file contains the following code:

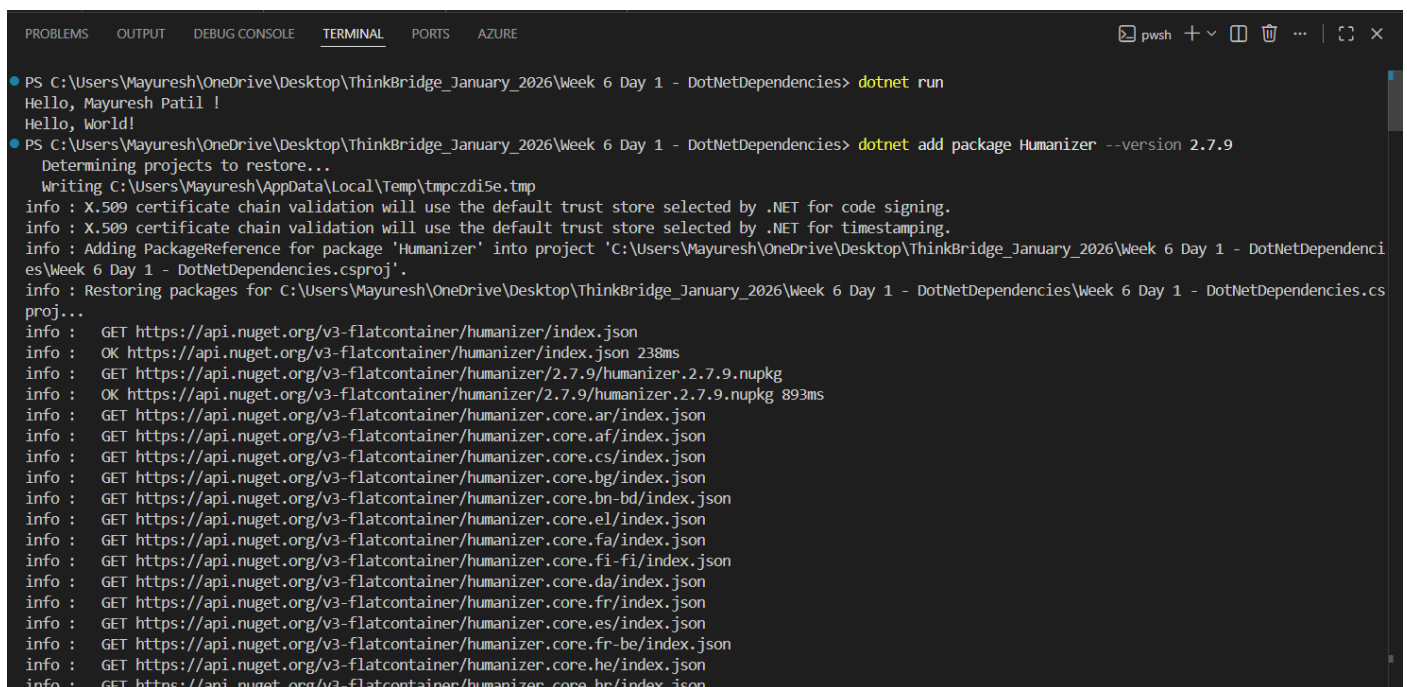
```
1 // See https://aka.ms/new-console-template for more information
2 Console.WriteLine("Hello, Mayuresh Patil !");
3
```

The Terminal pane at the bottom shows the output of the following commands:

```
PS C:\Users\Mayuresh\OneDrive\Desktop\ThinkBridge_January_2026\Week 6 Day 1 - DotNetDependencies> dotnet new console -f net8.0
The template "Console App" was created successfully.

Processing post-creation actions...
Restoring C:\Users\Mayuresh\OneDrive\Desktop\ThinkBridge_January_2026\Week 6 Day 1 - DotNetDependencies\Week 6 Day 1 - DotNetDependencies.csproj:
Determining projects to restore...
  Restored C:\Users\Mayuresh\OneDrive\Desktop\ThinkBridge_January_2026\Week 6 Day 1 - DotNetDependencies\Week 6 Day 1 - DotNetDependencies.csproj (in 61 ms).
Restore succeeded.

PS C:\Users\Mayuresh\OneDrive\Desktop\ThinkBridge_January_2026\Week 6 Day 1 - DotNetDependencies> dotnet run
Hello, World!
PS C:\Users\Mayuresh\OneDrive\Desktop\ThinkBridge_January_2026\Week 6 Day 1 - DotNetDependencies> dotnet run
Hello, World!
PS C:\Users\Mayuresh\OneDrive\Desktop\ThinkBridge_January_2026\Week 6 Day 1 - DotNetDependencies> dotnet run
Hello, Mayuresh Patil !
PS C:\Users\Mayuresh\OneDrive\Desktop\ThinkBridge_January_2026\Week 6 Day 1 - DotNetDependencies>
```



The screenshot shows the Visual Studio Code terminal with the following output:

```
PS C:\Users\Mayuresh\OneDrive\Desktop\ThinkBridge_January_2026\Week 6 Day 1 - DotNetDependencies> dotnet run
Hello, Mayuresh Patil !
Hello, World!
PS C:\Users\Mayuresh\OneDrive\Desktop\ThinkBridge_January_2026\Week 6 Day 1 - DotNetDependencies> dotnet add package Humanizer --version 2.7.9
Determining projects to restore...
Writing C:\Users\Mayuresh\AppData\Local\Temp\tmpczdi5e.tmp
info : X.509 certificate chain validation will use the default trust store selected by .NET for code signing.
info : X.509 certificate chain validation will use the default trust store selected by .NET for timestamping.
info : Adding PackageReference for package 'Humanizer' into project 'C:\Users\Mayuresh\OneDrive\Desktop\ThinkBridge_January_2026\Week 6 Day 1 - DotNetDependencies\Week 6 Day 1 - DotNetDependencies.csproj'.
info : Restoring packages for C:\Users\Mayuresh\OneDrive\Desktop\ThinkBridge_January_2026\Week 6 Day 1 - DotNetDependencies\Week 6 Day 1 - DotNetDependencies.csproj...
info :   GET https://api.nuget.org/v3-flatcontainer/humanizer/index.json
info :   OK https://api.nuget.org/v3-flatcontainer/humanizer/index.json 238ms
info :   GET https://api.nuget.org/v3-flatcontainer/humanizer/2.7.9/humanizer.2.7.9.nupkg
info :   OK https://api.nuget.org/v3-flatcontainer/humanizer/2.7.9/humanizer.2.7.9.nupkg 893ms
info :   GET https://api.nuget.org/v3-flatcontainer/humanizer.core.ar/index.json
info :   GET https://api.nuget.org/v3-flatcontainer/humanizer.core.af/index.json
info :   GET https://api.nuget.org/v3-flatcontainer/humanizer.core.cs/index.json
info :   GET https://api.nuget.org/v3-flatcontainer/humanizer.core.bg/index.json
info :   GET https://api.nuget.org/v3-flatcontainer/humanizer.core.bn-bd/index.json
info :   GET https://api.nuget.org/v3-flatcontainer/humanizer.core.el/index.json
info :   GET https://api.nuget.org/v3-flatcontainer/humanizer.core.fa/index.json
info :   GET https://api.nuget.org/v3-flatcontainer/humanizer.core.fi-fi/index.json
info :   GET https://api.nuget.org/v3-flatcontainer/humanizer.core.da/index.json
info :   GET https://api.nuget.org/v3-flatcontainer/humanizer.core.fr/index.json
info :   GET https://api.nuget.org/v3-flatcontainer/humanizer.core.es/index.json
info :   GET https://api.nuget.org/v3-flatcontainer/humanizer.core.fr-be/index.json
info :   GET https://api.nuget.org/v3-flatcontainer/humanizer.core.he/index.json
info :   GET https://api.nuget.org/v3-flatcontainer/humanizer.core.hr/index.json
```

```

PS C:\Users\Mayuresh\OneDrive\Desktop\ThinkBridge_January_2026\Week 6 Day 1 - DotNetDependencies> dotnet add package Humanizer --version 2.7.9
info : Installed Humanizer.Core.pt 2.7.9 from https://api.nuget.org/v3/index.json to C:\Users\Mayuresh\.nuget\packages\humanizer.core.pt\2.7.9 with content hash
M5/hzNDPJD/Omp5S30LqBwukj8WmxASU0XIK9BEpYOYuegsp8W4RaCvj79BNMwKjyBjDjHmq1f4E2UvtCgg==.
info : Installed Humanizer.Core.ms-MY 2.7.9 from https://api.nuget.org/v3/index.json to C:\Users\Mayuresh\.nuget\packages\humanizer.core.ms-my\2.7.9 with conten
t hash 609Cd0239ddC2inn32ialw1BEkBOHoBoHhkaw8x114SlnYEDRaS9a+GRhJZ0d6s3H3IUznbk3qv7sSqC8XI67g==.
info : Installed Humanizer.Core.nb-NO 2.7.9 from https://api.nuget.org/v3/index.json to C:\Users\Mayuresh\.nuget\packages\humanizer.core.nb-no\2.7.9 with conten
t hash EpDl9LlyWNC+/Kk1f4cA8goiFgrSQ2evpMYRzUCJCSSmASf2OR7owkGiwWtB3nNS/SRkrC2ouQBwiXPj8bkQhg==.
info : Installed Humanizer.Core.pl 2.7.9 from https://api.nuget.org/v3/index.json to C:\Users\Mayuresh\.nuget\packages\humanizer.core.pl\2.7.9 with content hash
SNMNRp3GklOpVFWFXEMaUggCR27rVbD2z1CPmSVL+2RLIHcdrdNbrxRX8neKxz11Enmue550Gg9ipVRTrZmf3Q==.
info : GET https://api.nuget.org/v3/vulnerabilities/index.json
info : OK https://api.nuget.org/v3/vulnerabilities/index.json 235ms
info : GET https://api.nuget.org/v3-vulnerabilities/2026.02.21.05.31.43/vulnerability.base.json
info : GET https://api.nuget.org/v3-vulnerabilities/2026.02.21.05.31.43/2026.02.23.23.31.51/vulnerability.update.json
info : OK https://api.nuget.org/v3-vulnerabilities/2026.02.21.05.31.43/vulnerability.base.json 236ms
info : OK https://api.nuget.org/v3-vulnerabilities/2026.02.21.05.31.43/2026.02.23.23.31.51/vulnerability.update.json 250ms
info : Package 'Humanizer' is compatible with all the specified frameworks in project 'C:\Users\Mayuresh\OneDrive\Desktop\ThinkBridge_January_2026\Week 6 Day 1
- DotNetDependencies\Week 6 Day 1 - DotNetDependencies.csproj'.
info : PackageReference for package 'Humanizer' version '2.7.9' added to file 'C:\Users\Mayuresh\OneDrive\Desktop\ThinkBridge_January_2026\Week 6 Day 1 - DotNet
Dependencies\Week 6 Day 1 - DotNetDependencies.csproj'.
info : Writing assets file to disk. Path: C:\Users\Mayuresh\OneDrive\Desktop\ThinkBridge_January_2026\Week 6 Day 1 - DotNetDependencies\obj\project.assets.json
log : Restored C:\Users\Mayuresh\OneDrive\Desktop\ThinkBridge_January_2026\Week 6 Day 1 - DotNetDependencies\Week 6 Day 1 - DotNetDependencies.csproj (in 4.47
sec).
PS C:\Users\Mayuresh\OneDrive\Desktop\ThinkBridge_January_2026\Week 6 Day 1 - DotNetDependencies>

```

The screenshot shows the Visual Studio IDE with the following components:

- Explorer:** Shows the project structure with folders `bin`, `obj`, and `Program.cs`. The file `Week 6 Day 1 - DotNetDependencies.sln` is selected.
- Program.cs:** Contains the following code:


```

7 //Add a NuGet package by using the .NET Core tool
8 Console.WriteLine("Quantities:");
9 HumanizeQuantities();
10
11 Console.WriteLine("\nDate/Time Manipulation:");
12 HumanizeDates();
13
14 static void HumanizeQuantities()
15 {
16     Console.WriteLine("case".ToQuantity(0));
17     Console.WriteLine("case".ToQuantity(1));
18     Console.WriteLine("case".ToQuantity(5));
19 }
20
21 static void HumanizeDates()
22 {
23     Console.WriteLine(DateTime.UtcNow.AddHours(-24).Humanize());
24     Console.WriteLine(DateTime.UtcNow.AddHours(-2).Humanize());
25     Console.WriteLine(TimeSpan.FromDays(1).Humanize());
26     Console.WriteLine(TimeSpan.FromDays(16).Humanize());
27 }

```
- Terminal:** Shows the output of the `dotnet run` command:

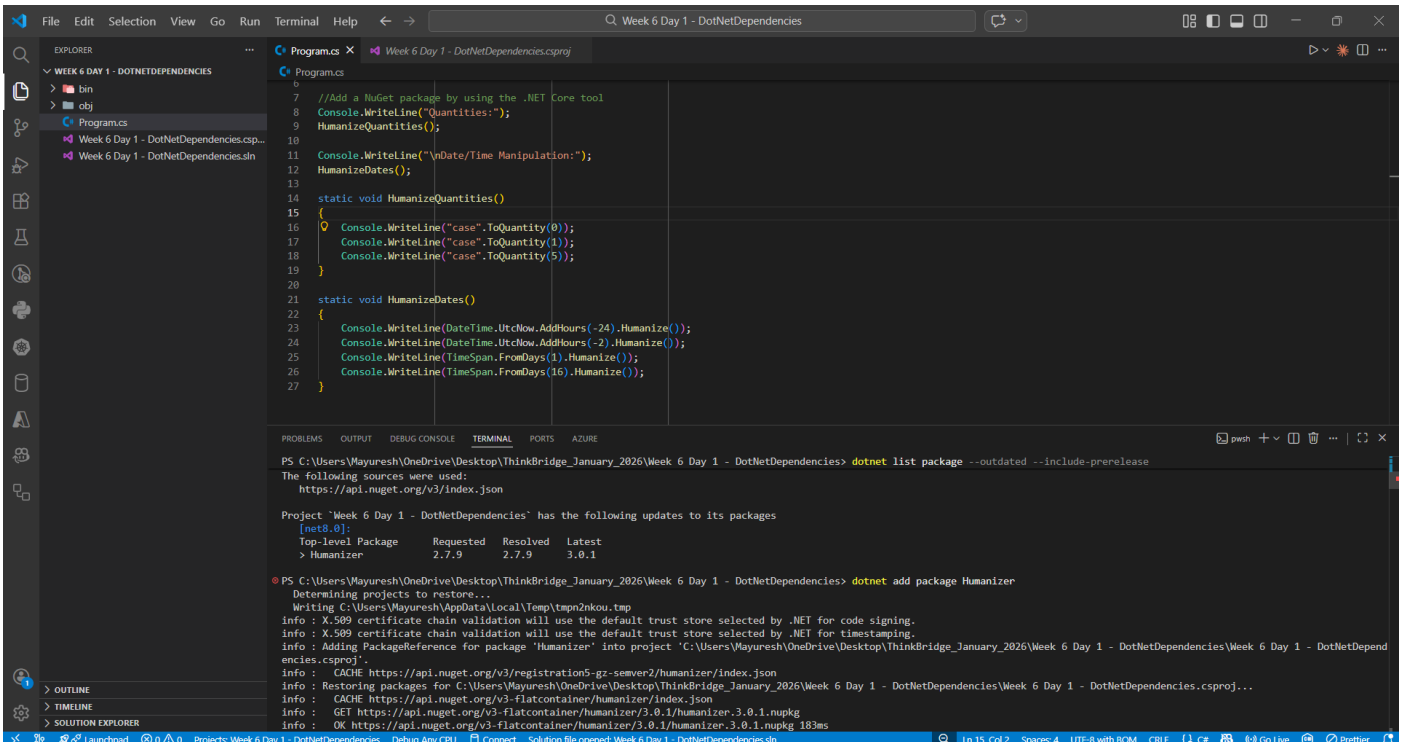

```

PS C:\Users\Mayuresh\OneDrive\Desktop\ThinkBridge_January_2026\Week 6 Day 1 - DotNetDependencies> dotnet run
Quantities:
0 cases
1 case
5 cases

Date/Time Manipulation:
yesterday
2 hours ago
1 day
2 weeks

```
- Output:** Shows the logs from the NuGet package manager, including the installation of `Humanizer` and the restoration of the project.

Exercise - Manage dependency updates in your .NET project



```
7 //Add a NuGet package by using the .NET Core tool
8 Console.WriteLine("Quantities:");
9 HumanizeQuantities();
10
11 Console.WriteLine("\nDate/Time Manipulation:");
12 HumanizeDates();
13
14
15 static void HumanizeQuantities()
16 {
17     Console.WriteLine("case".ToQuantity(0));
18     Console.WriteLine("case".ToQuantity(1));
19     Console.WriteLine("case".ToQuantity(5));
20 }
21
22 static void HumanizeDates()
23 {
24     Console.WriteLine(DateTime.UtcNow.AddHours(-24).Humanize());
25     Console.WriteLine(DateTime.UtcNow.AddHours(-2).Humanize());
26     Console.WriteLine(TimeSpan.FromDays(1).Humanize());
27     Console.WriteLine(TimeSpan.FromDays(16).Humanize());
28 }
```

PS C:\Users\Mayuresh\OneDrive\Desktop\ThinkBridge_January_2026\Week 6 Day 1 - DotNetDependencies> dotnet list package --outdated --include-prerelease

The following sources were used:

https://api.nuget.org/v3/index.json

Project 'Week 6 Day 1 - DotNetDependencies' has the following updates to its packages

Package	Requested	Resolved	Latest
Humanizer	2.7.9	2.7.9	3.0.1

PS C:\Users\Mayuresh\OneDrive\Desktop\ThinkBridge_January_2026\Week 6 Day 1 - DotNetDependencies> dotnet add package Humanizer

Determining projects to restore...

Writing C:\Users\Mayuresh\AppData\Local\Temp\tmpn2nkou.tmp

info : X.509 certificate chain validation will use the default trust store selected by .NET for code signing.

info : X.509 certificate chain validation will use the default trust store selected by .NET for timestamping.

info : Adding PackageReference for package 'Humanizer' into project 'C:\Users\Mayuresh\OneDrive\Desktop\ThinkBridge_January_2026\Week 6 Day 1 - DotNetDependencies\Week 6 Day 1 - DotNetDependencies.cspj'.

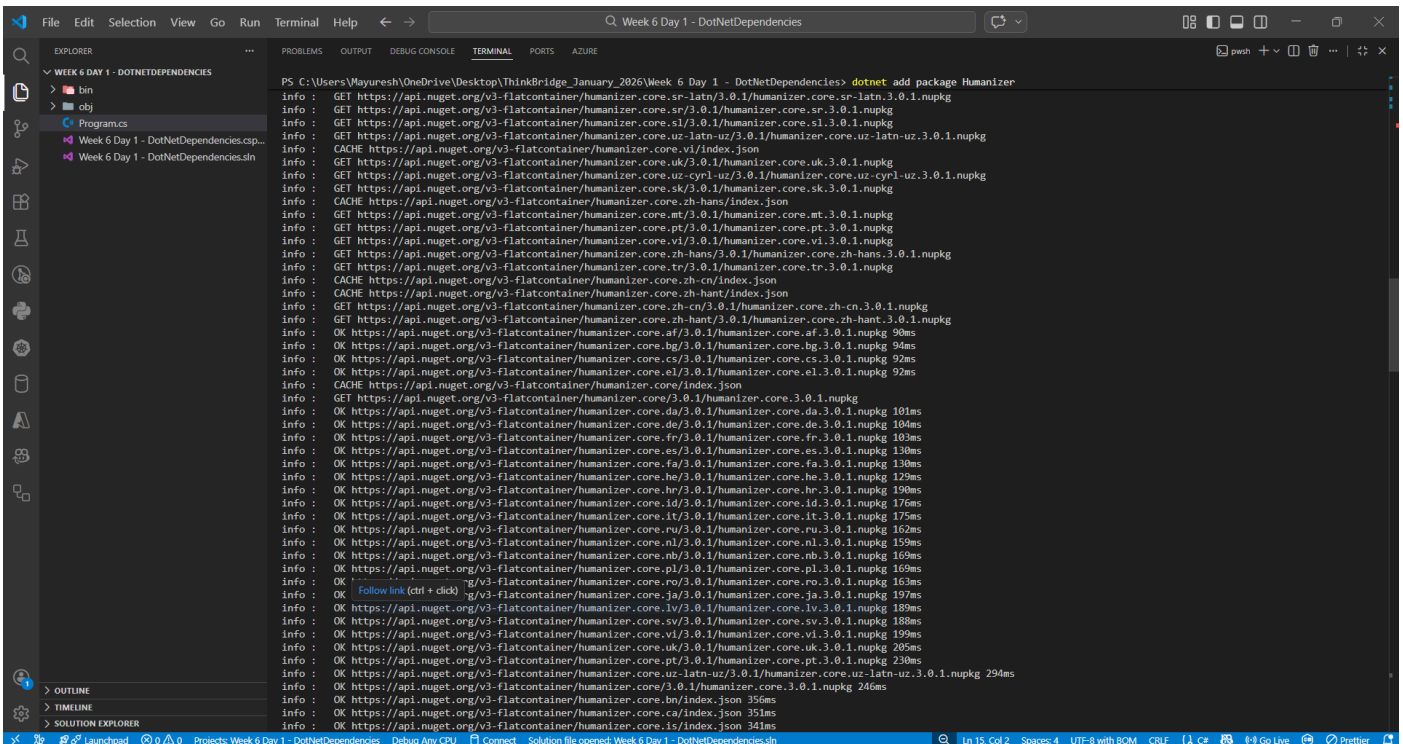
info : CACHE https://api.nuget.org/v3/registrations-gz-sserver2/humanizer/index.json

info : Restoring packages for C:\Users\Mayuresh\OneDrive\Desktop\ThinkBridge_January_2026\Week 6 Day 1 - DotNetDependencies\Week 6 Day 1 - DotNetDependencies.cspj...

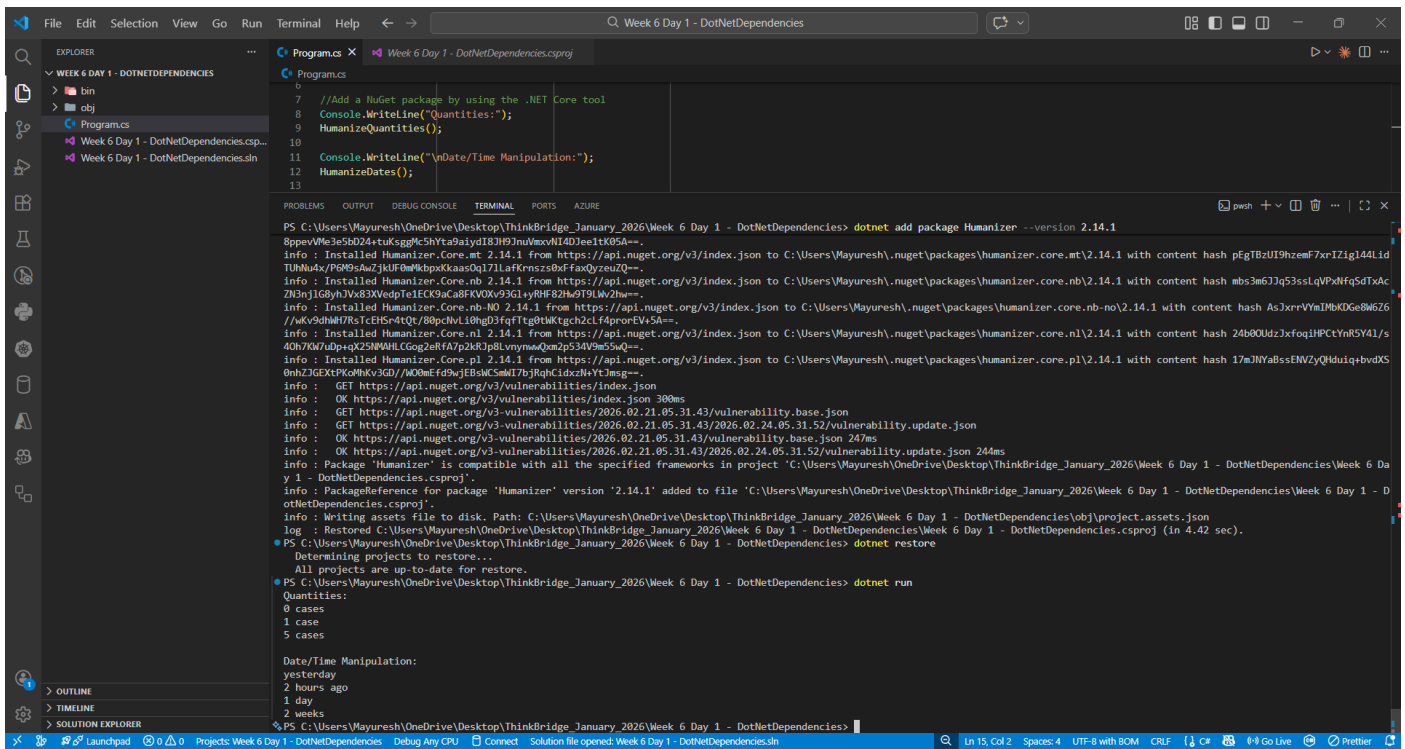
info : CACHE https://api.nuget.org/v3-flatcontainer/humanizer/index.json

info : GET https://api.nuget.org/v3-flatcontainer/humanizer/3.0.1/humanizer.3.0.1.nupkg

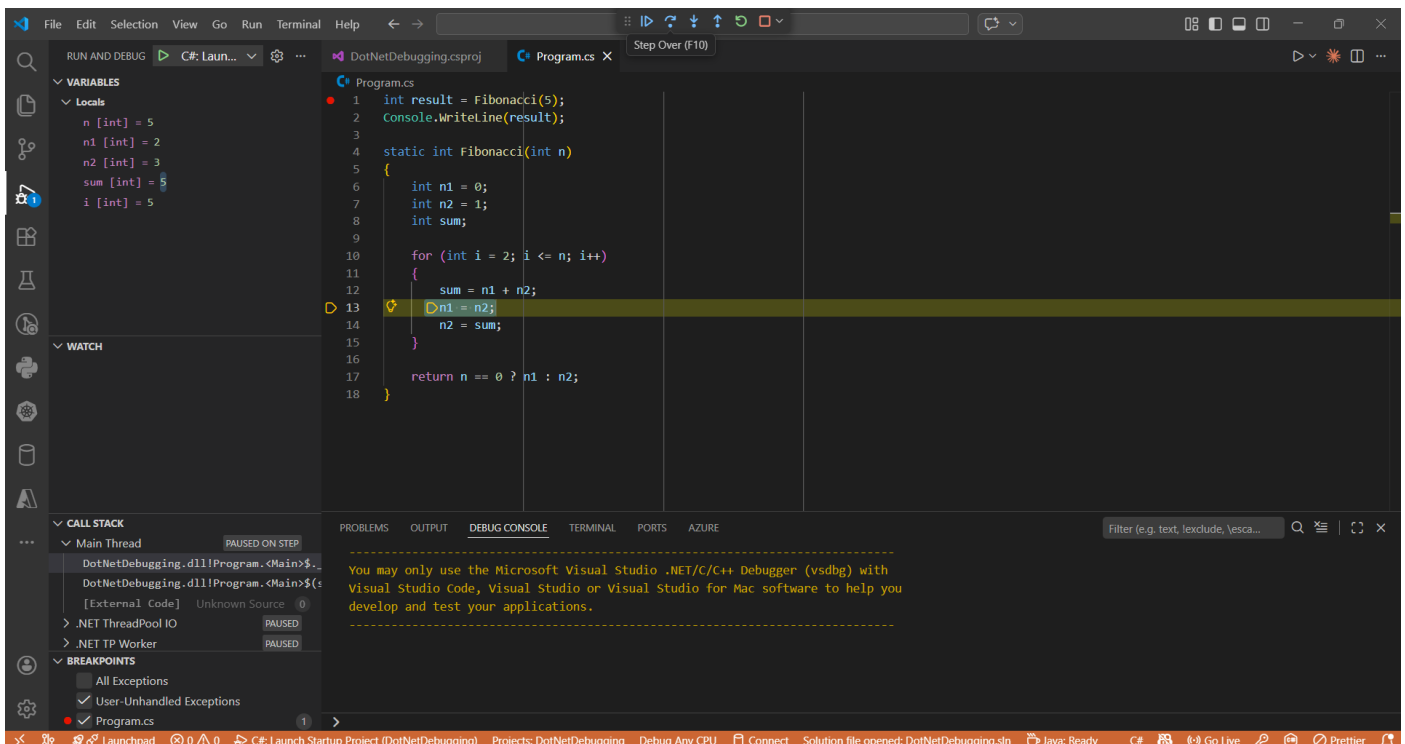
info : OK https://api.nuget.org/v3-flatcontainer/humanizer/3.0.1/humanizer.3.0.1.nupkg 183ms

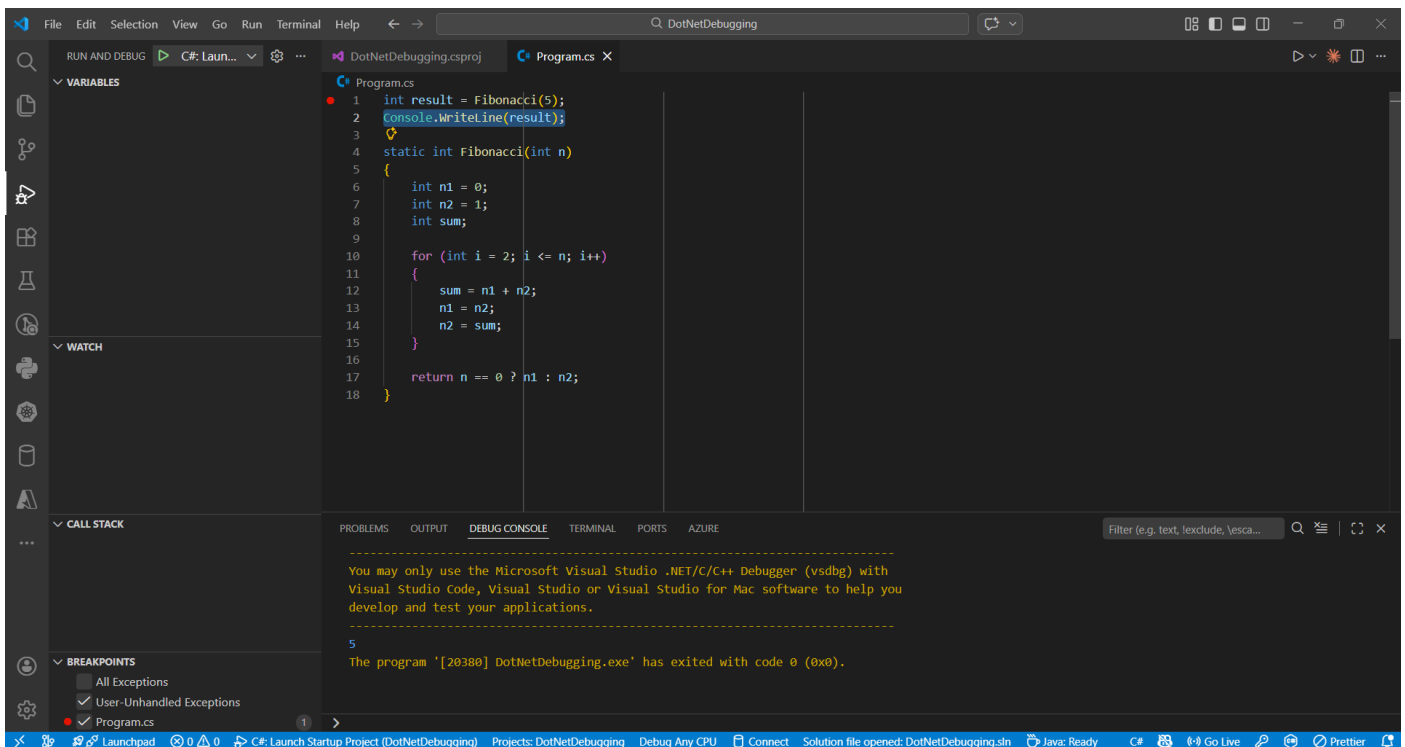
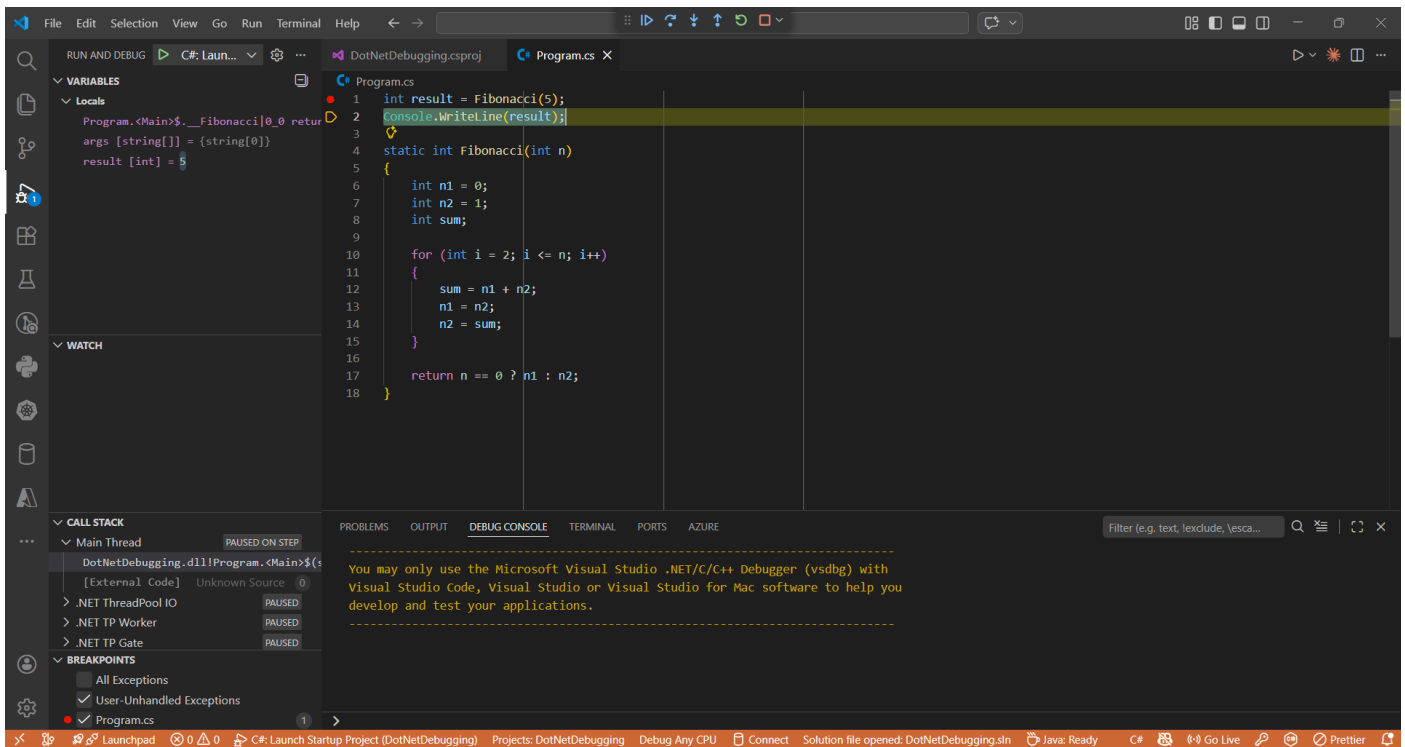


```
PS C:\Users\Mayuresh\OneDrive\Desktop\ThinkBridge_January_2026\Week 6 Day 1 - DotNetDependencies> dotnet add package Humanizer
info : GET https://api.nuget.org/v3-flatcontainer/humanizer.core.sr-latn/3.0.1/humanizer.core.sr-latn.3.0.1.nupkg
info : GET https://api.nuget.org/v3-flatcontainer/humanizer.core.sr/3.0.1/humanizer.core.sr.3.0.1.nupkg
info : GET https://api.nuget.org/v3-flatcontainer/humanizer.core.sr-latn-uz/3.0.1/humanizer.core.sr-latn-uz.3.0.1.nupkg
info : GET https://api.nuget.org/v3-flatcontainer/humanizer.core.sr-latn-uz/3.0.1/humanizer.core.sr-latn-uz.3.0.1.nupkg
info : CACHE https://api.nuget.org/v3-flatcontainer/humanizer.core.vi/index.json
info : GET https://api.nuget.org/v3-flatcontainer/humanizer.core.uk/3.0.1/humanizer.core.uk.3.0.1.nupkg
info : GET https://api.nuget.org/v3-flatcontainer/humanizer.core.uz-cyrl-uz/3.0.1/humanizer.core.uz-cyrl-uz.3.0.1.nupkg
info : GET https://api.nuget.org/v3-flatcontainer/humanizer.core.sr/3.0.1/humanizer.core.sr.3.0.1.nupkg
info : CACHE https://api.nuget.org/v3-flatcontainer/humanizer.core.sk-hans/index.json
info : GET https://api.nuget.org/v3-flatcontainer/humanizer.core.mt/3.0.1/humanizer.core.mt.3.0.1.nupkg
info : GET https://api.nuget.org/v3-flatcontainer/humanizer.core.pt/3.0.1/humanizer.core.pt.3.0.1.nupkg
info : GET https://api.nuget.org/v3-flatcontainer/humanizer.core.vi/3.0.1/humanizer.core.vi.3.0.1.nupkg
info : GET https://api.nuget.org/v3-flatcontainer/humanizer.core.zh-hans/3.0.1/humanizer.core.zh-hans.3.0.1.nupkg
info : GET https://api.nuget.org/v3-flatcontainer/humanizer.core.tr/3.0.1/humanizer.core.tr.3.0.1.nupkg
info : CACHE https://api.nuget.org/v3-flatcontainer/humanizer.core.zh-cn/index.json
info : CACHE https://api.nuget.org/v3-flatcontainer/humanizer.core.zh-hant/index.json
info : GET https://api.nuget.org/v3-flatcontainer/humanizer.core.zh-cn/3.0.1/humanizer.core.zh-cn.3.0.1.nupkg
info : GET https://api.nuget.org/v3-flatcontainer/humanizer.core.zh-hant/3.0.1/humanizer.core.zh-hant.3.0.1.nupkg
info : OK https://api.nuget.org/v3-flatcontainer/humanizer.core.af/3.0.1/humanizer.core.af.3.0.1.nupkg 90ms
info : OK https://api.nuget.org/v3-flatcontainer/humanizer.core.bg/3.0.1/humanizer.core.bg.3.0.1.nupkg 94ms
info : OK https://api.nuget.org/v3-flatcontainer/humanizer.core.cs/3.0.1/humanizer.core.cs.3.0.1.nupkg 92ms
info : OK https://api.nuget.org/v3-flatcontainer/humanizer.core.el/3.0.1/humanizer.core.el.3.0.1.nupkg 92ms
info : CACHE https://api.nuget.org/v3-flatcontainer/humanizer.core/index.json
info : GET https://api.nuget.org/v3-flatcontainer/humanizer.core/3.0.1/humanizer.core.3.0.1.nupkg
info : OK https://api.nuget.org/v3-flatcontainer/humanizer.core.da/3.0.1/humanizer.core.da.3.0.1.nupkg 101ms
info : OK https://api.nuget.org/v3-flatcontainer/humanizer.core.de/3.0.1/humanizer.core.de.3.0.1.nupkg 104ms
info : OK https://api.nuget.org/v3-flatcontainer/humanizer.core.fr/3.0.1/humanizer.core.fr.3.0.1.nupkg 103ms
info : OK https://api.nuget.org/v3-flatcontainer/humanizer.core.es/3.0.1/humanizer.core.es.3.0.1.nupkg 130ms
info : OK https://api.nuget.org/v3-flatcontainer/humanizer.core.fa/3.0.1/humanizer.core.fa.3.0.1.nupkg 130ms
info : OK https://api.nuget.org/v3-flatcontainer/humanizer.core.he/3.0.1/humanizer.core.he.3.0.1.nupkg 129ms
info : OK https://api.nuget.org/v3-flatcontainer/humanizer.core.hr/3.0.1/humanizer.core.hr.3.0.1.nupkg 190ms
info : OK https://api.nuget.org/v3-flatcontainer/humanizer.core.id/3.0.1/humanizer.core.id.3.0.1.nupkg 176ms
info : OK https://api.nuget.org/v3-flatcontainer/humanizer.core.it/3.0.1/humanizer.core.it.3.0.1.nupkg 175ms
info : OK https://api.nuget.org/v3-flatcontainer/humanizer.core.ru/3.0.1/humanizer.core.ru.3.0.1.nupkg 162ms
info : OK https://api.nuget.org/v3-flatcontainer/humanizer.core.nl/3.0.1/humanizer.core.nl.3.0.1.nupkg 159ms
info : OK https://api.nuget.org/v3-flatcontainer/humanizer.core.nb/3.0.1/humanizer.core.nb.3.0.1.nupkg 169ms
info : OK https://api.nuget.org/v3-flatcontainer/humanizer.core.pl/3.0.1/humanizer.core.pl.3.0.1.nupkg 169ms
info : OK https://api.nuget.org/v3-flatcontainer/humanizer.core.ro/3.0.1/humanizer.core.ro.3.0.1.nupkg 163ms
info : OK https://api.nuget.org/v3-flatcontainer/humanizer.core.ja/3.0.1/humanizer.core.ja.3.0.1.nupkg 197ms
info : OK https://api.nuget.org/v3-flatcontainer/humanizer.core.lv/3.0.1/humanizer.core.lv.3.0.1.nupkg 189ms
info : OK https://api.nuget.org/v3-flatcontainer/humanizer.core.sv/3.0.1/humanizer.core.sv.3.0.1.nupkg 188ms
info : OK https://api.nuget.org/v3-flatcontainer/humanizer.core.vi/3.0.1/humanizer.core.vi.3.0.1.nupkg 199ms
info : OK https://api.nuget.org/v3-flatcontainer/humanizer.core.uk/3.0.1/humanizer.core.uk.3.0.1.nupkg 205ms
info : OK https://api.nuget.org/v3-flatcontainer/humanizer.core.pt/3.0.1/humanizer.core.pt.3.0.1.nupkg 230ms
info : OK https://api.nuget.org/v3-flatcontainer/humanizer.core.uz-latn-uz/3.0.1/humanizer.core.uz-latn-uz.3.0.1.nupkg 294ms
info : OK https://api.nuget.org/v3-flatcontainer/humanizer.core/3.0.1/humanizer.core.3.0.1.nupkg 246ms
info : OK https://api.nuget.org/v3-flatcontainer/humanizer.core.bn/index.json 35ms
info : OK https://api.nuget.org/v3-flatcontainer/humanizer.core.ca/index.json 351ms
info : OK https://api.nuget.org/v3-flatcontainer/humanizer.core.is/index.json 341ms
```



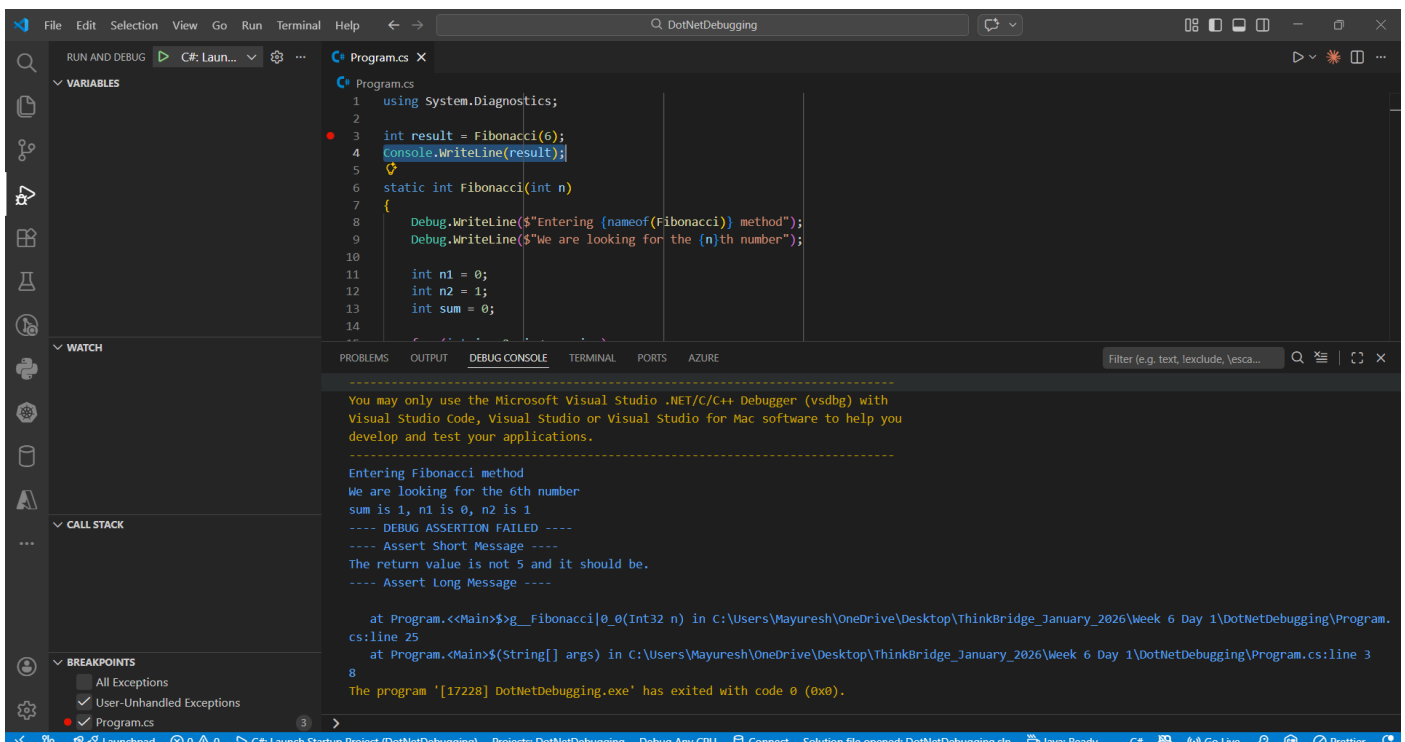
Exercise - Debug with Visual Studio Code



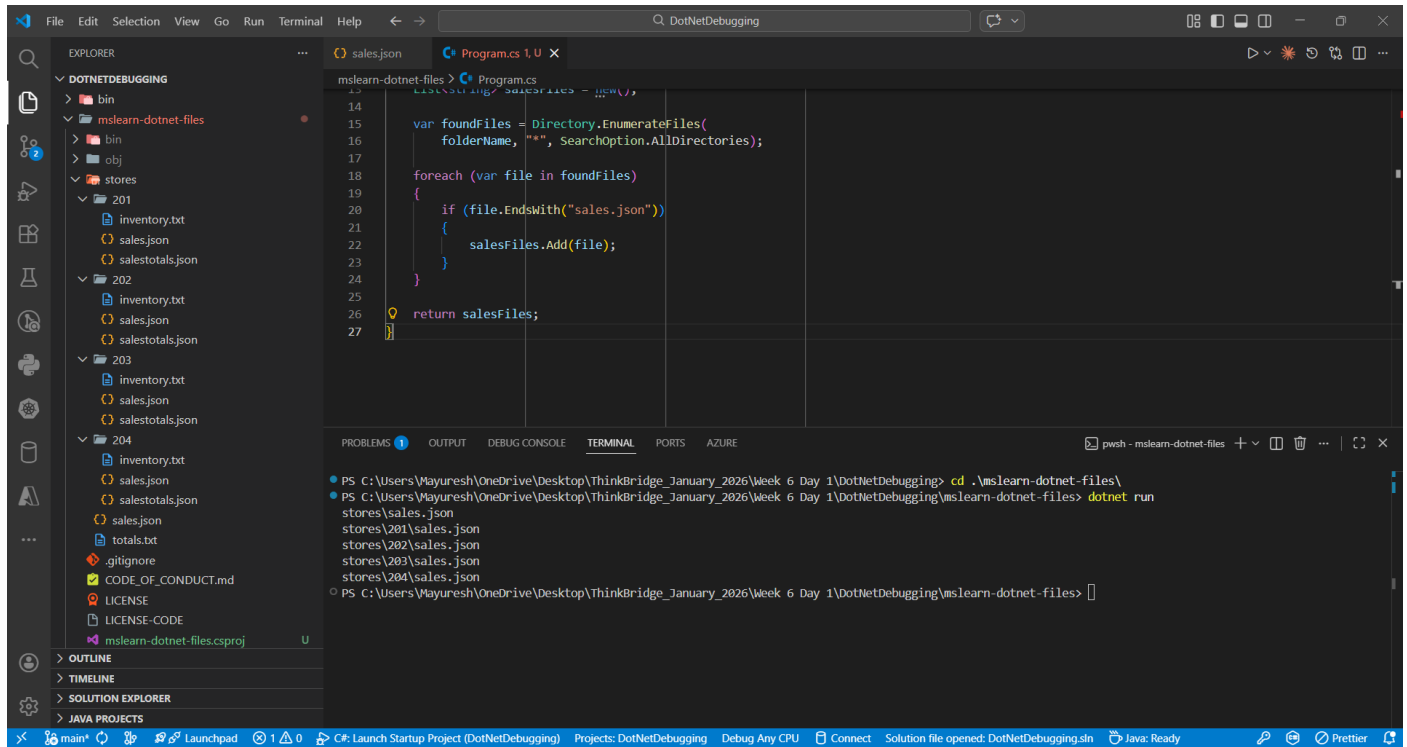


Exercise - Logging and tracing

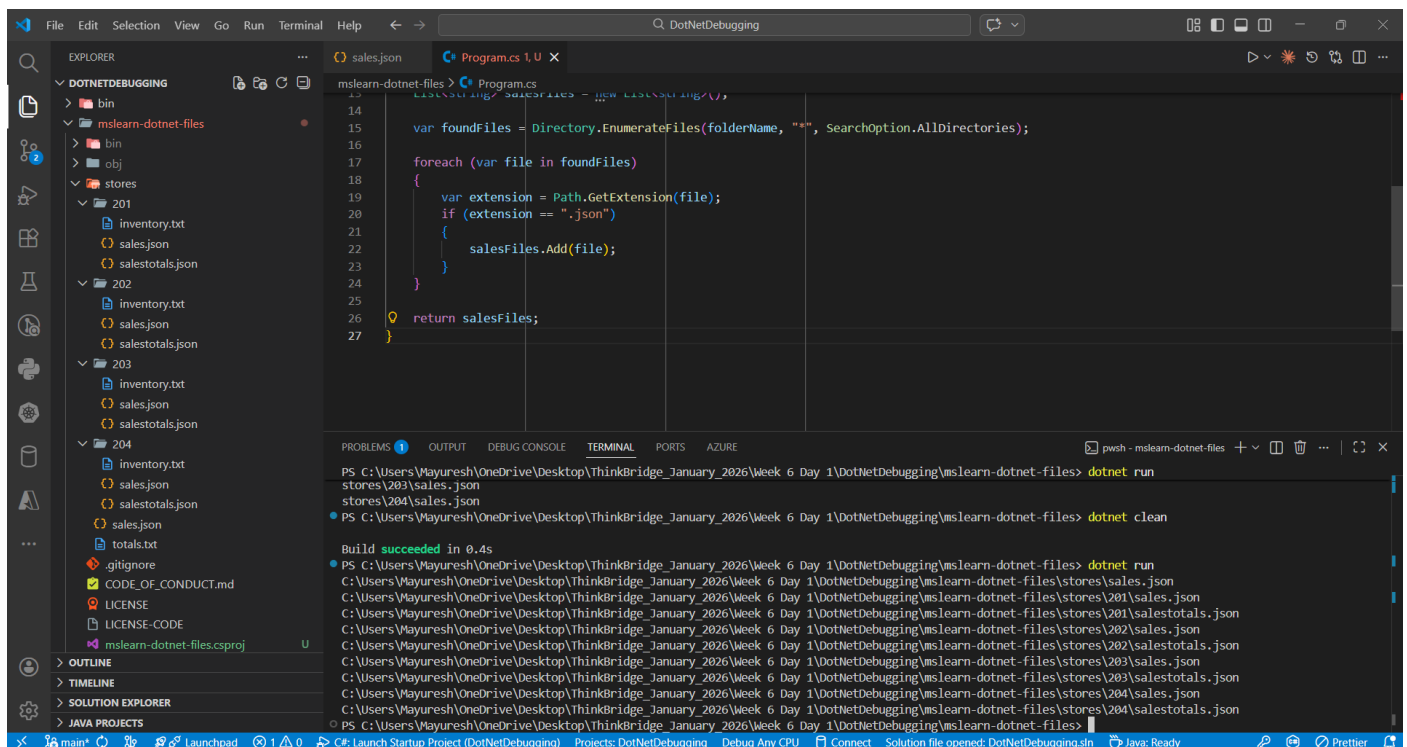
```
Program.cs X
Program.cs
1 using System.Diagnostics;
2
3 int result = Fibonacci(6);
4 Console.WriteLine(result);
5
6 static int Fibonacci(int n)
7 {
8     Debug.WriteLine($"Entering {nameof(Fibonacci)} method");
9     Debug.WriteLine($"We are looking for the {n}th number");
10
11     int n1 = 0;
12     int n2 = 1;
13     int sum = 0;
14
15     for (int i = 2; i <= n; i++)
16     {
17         sum = n1 + n2;
18
19         Debug.WriteLineIf(sum == 1, $"sum is 1, n1 is {n1}, n2 is {n2}");
20
21         n1 = n2;
22         n2 = sum;
23     }
24
25     Debug.Assert(n2 == 5, "The return value is not 5 and it should be.");
26
27     return n == 0 ? n1 : n2;
28 }
```



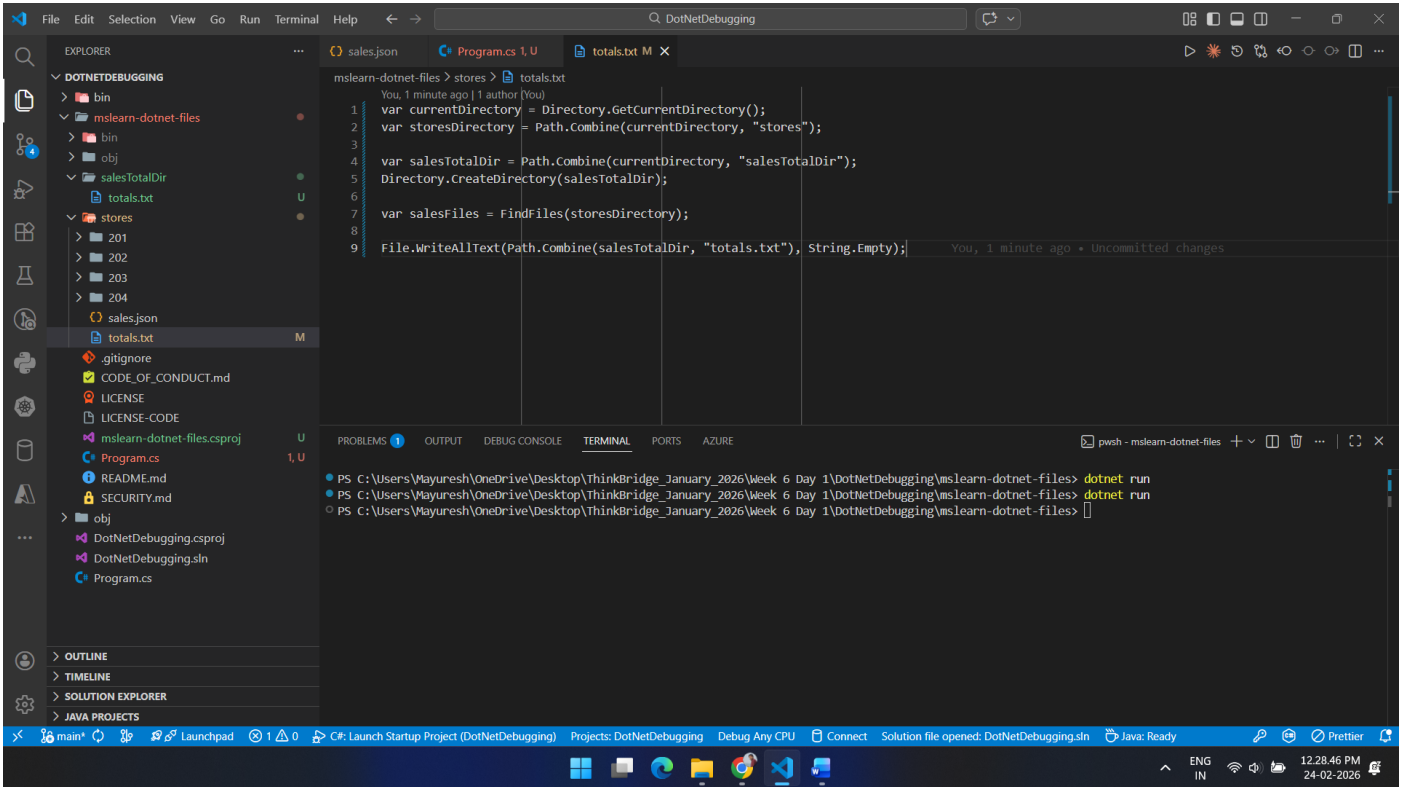
Exercise - Work with the file system



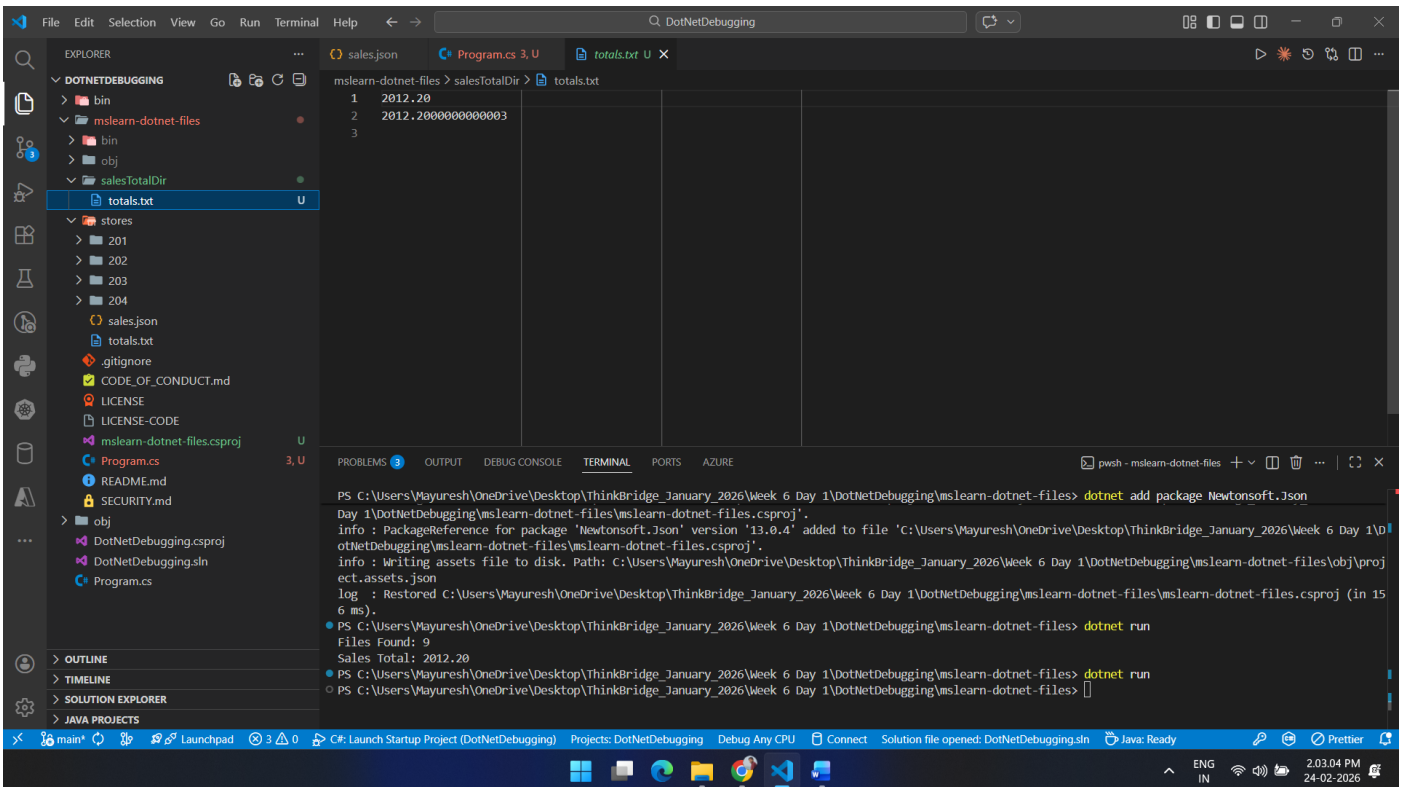
Exercise - Work with paths



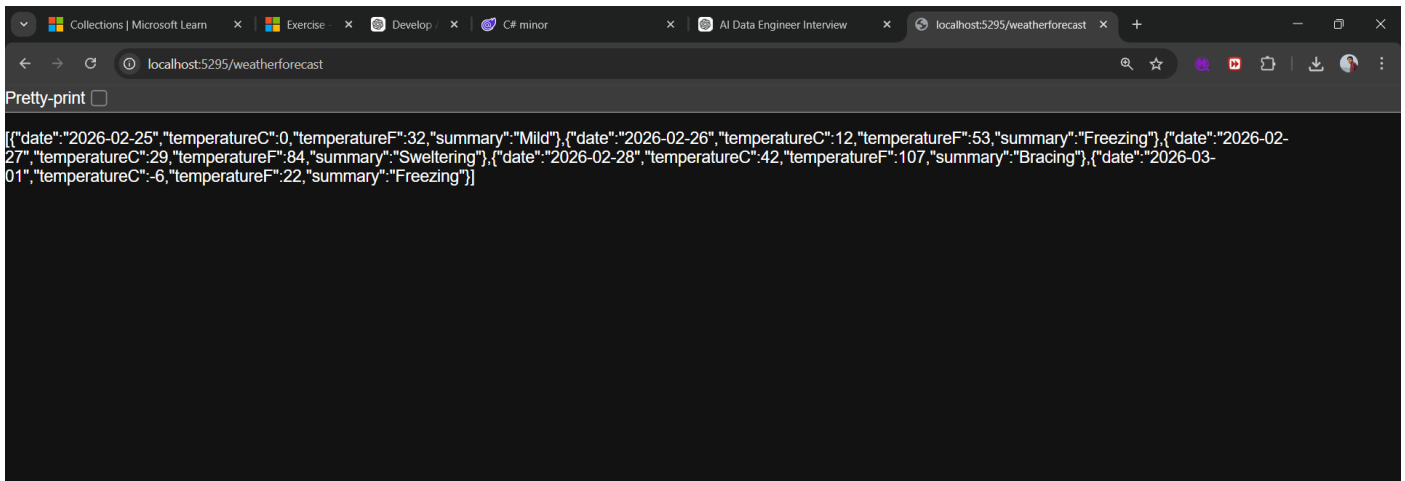
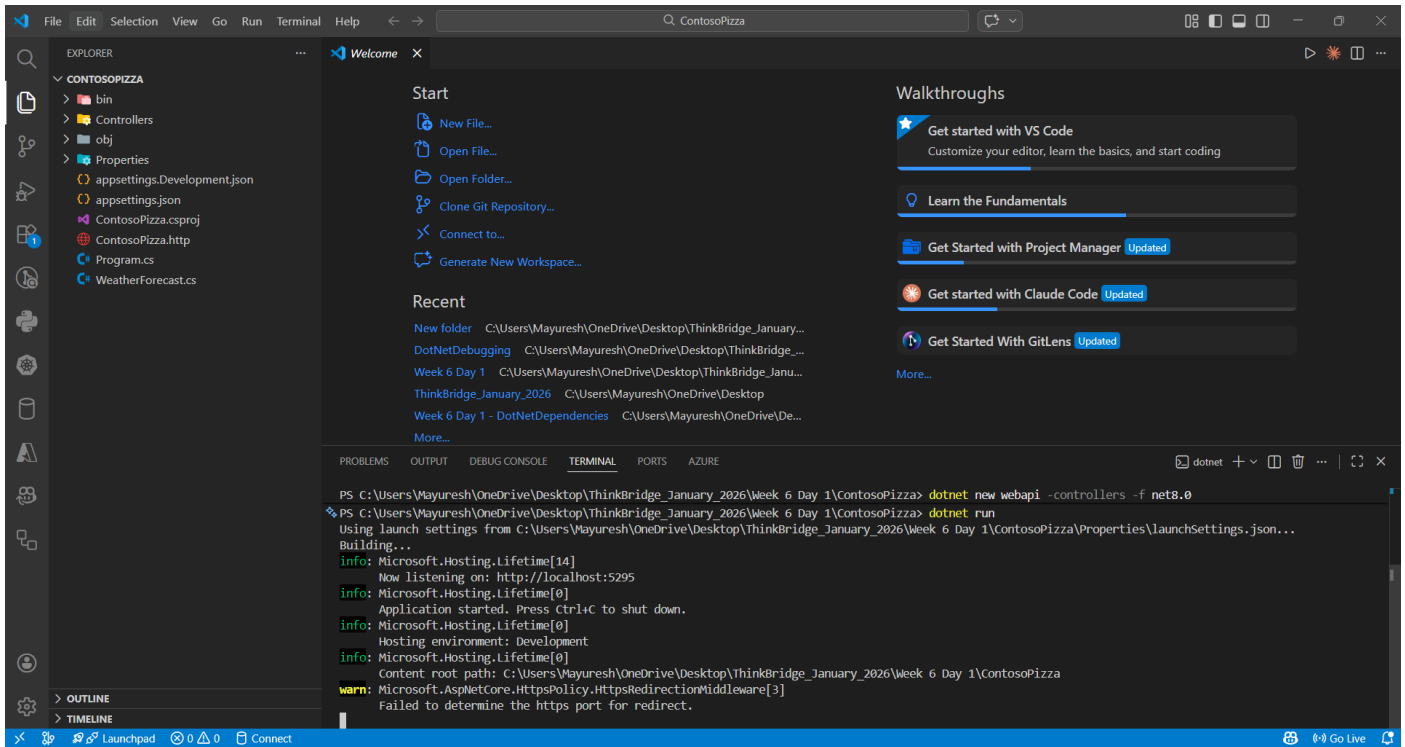
Exercise - Create files and directories



Exercise - Read and write to files



Exercise - Create a web API project



```
HTTP/1.1 200 OK
Connection: close
Content-Type: application/json; charset=utf-8
Date: Tue, 24 Feb 2026 08:41:10 GMT
Server: Kestrel
Transfer-Encoding: chunked
```

```
[
  {
    "date": "2026-02-25",
    "temperatureC": 49,
    "temperatureF": 120,
    "summary": "Cool"
  },
  {
    "date": "2026-02-26",
    "temperatureC": 36,
    "temperatureF": 96,
    "summary": "Bracing"
  },
  {
    "date": "2026-02-27",
    "temperatureC": 43,
    "temperatureF": 109,
    "summary": "Freezing"
  },
  {
    "date": "2026-02-28",
    "temperatureC": -19,
    "temperatureF": -2,
    "summary": "Hot"
  },
  {
    "date": "2026-03-01",
    "temperatureC": 19,
    "temperatureF": 66,
    "summary": "Bracing"
  }
]
```

The screenshot shows the Visual Studio Code interface with the 'ContosoPizza' project open. The Explorer pane on the left shows the project structure, including 'bin', 'Controllers', 'obj', 'Properties', 'appsettings.json', 'appsettings.json', 'ContosoPizza.csproj', 'ContosoPizza.http', 'ContosoPizza.sln', 'Program.cs', and 'WeatherForecast.cs'. The 'ContosoPizza.http' file is selected, showing a REST client with a 'GET' method for 'http://localhost:5295/WeatherForecast'. The terminal pane at the bottom shows the command 'httprepl http://localhost:5295' being executed, and the output of the REST client request, which is a JSON array of weather forecast data.

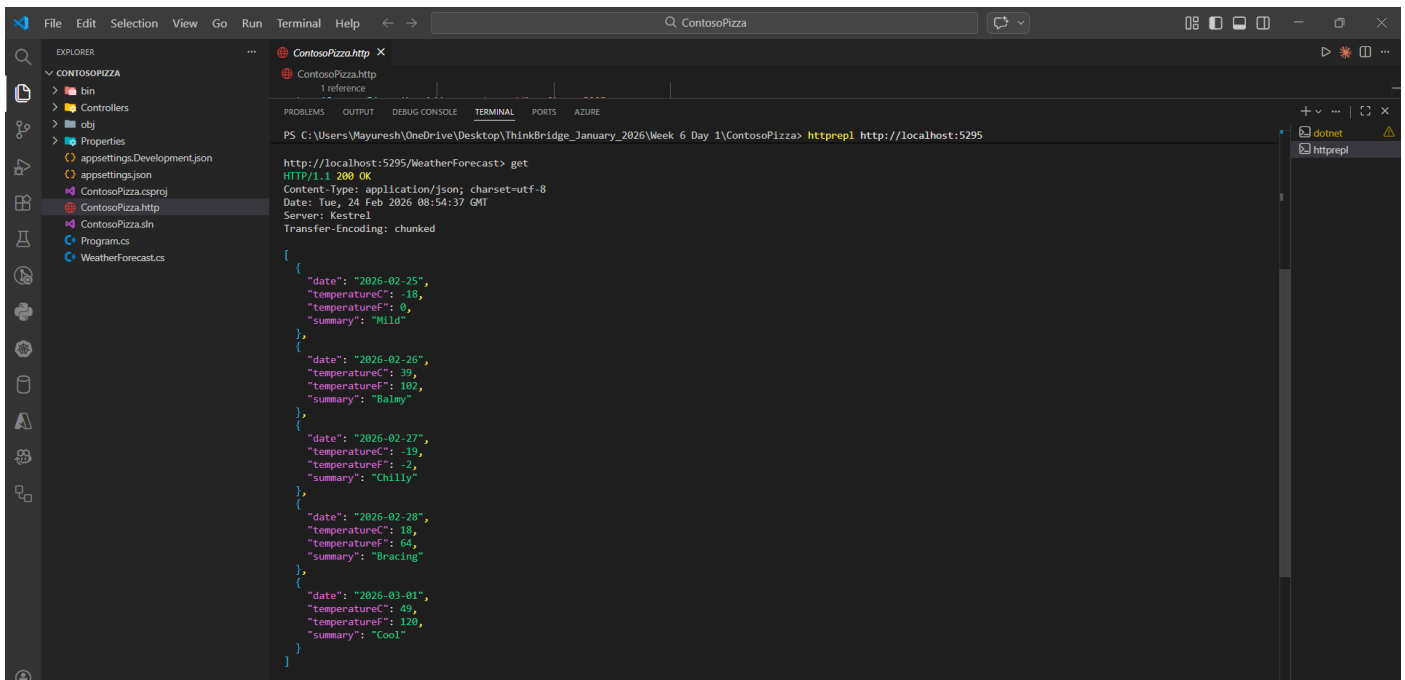
```
PS C:\Users\Mayuresh\OneDrive\Desktop\ThinkBridge_January_2026\Week 6 Day 1\ContosoPizza> httprepl http://localhost:5295
(Disconnected)> connect http://localhost:5295/
Using a base address of http://localhost:5295/
Using OpenAPI description at http://localhost:5295/swagger/v1/swagger.json
For detailed tool info, see https://aka.ms/http-repl-doc

http://localhost:5295/> ls
-
WeatherForecast [GET]

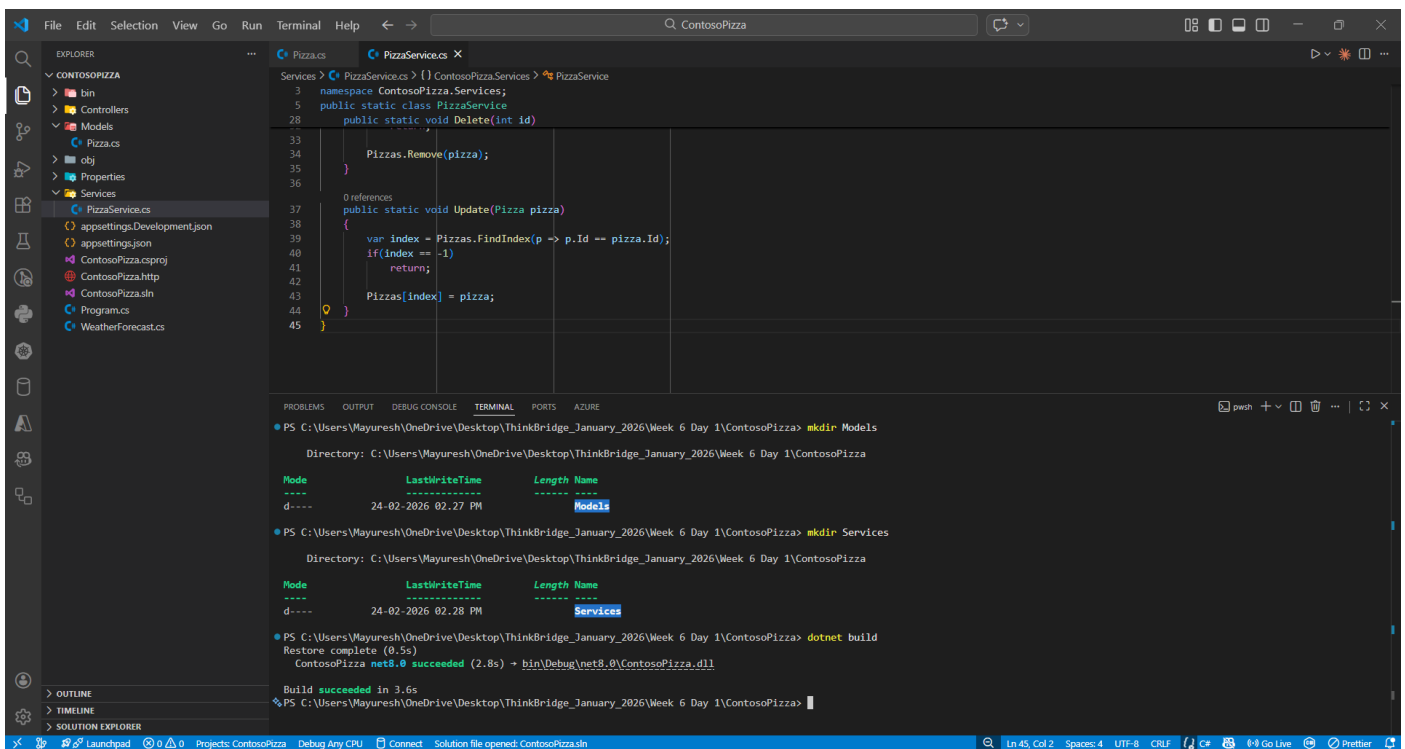
http://localhost:5295/> cd WeatherForecast
/WeatherForecast [GET]

http://localhost:5295/WeatherForecast> get
HTTP/1.1 200 OK
Content-Type: application/json; charset=utf-8
Date: Tue, 24 Feb 2026 08:54:37 GMT
Server: Kestrel
Transfer-Encoding: chunked

[
  {
    "date": "2026-02-25",
    "temperatureC": -18,
    "temperatureF": 0,
    "summary": "Mild"
  },
  {
    "date": "2026-02-26",
    "temperatureC": 39,
    "temperatureF": 102,
    "summary": "Balmy"
  },
  {
    "date": "2026-02-27",
    "temperatureC": -19,
    "temperatureF": -2,
    "summary": "Chilly"
  },
  {
    "date": "2026-02-28",
    "temperatureC": 18,
    "temperatureF": 64,
    "summary": "Bracing"
  }
]
```



Exercise - Add a data store



Exercise - Add a controller

The screenshot shows the Visual Studio IDE with the following components:

- EXPLORER:** The project structure for 'CONTOSOPIZZA' is visible, including 'bin', 'Controllers' (with 'PizzaController.cs'), 'Models', 'obj', 'Properties', 'Services' (with 'PizzaService.cs'), 'appsettings.Development.json', 'appsettings.json', 'ContosoPizza.csproj', 'ContosoPizza.http', 'ContosoPizza.sln', 'Program.cs', and 'WeatherForecast.cs'.
- EDITOR:** The 'ContosoPizza.http' file is open, showing a REST client request:

```
1 GET /pizza/
2 @ContosoPizza_HostAddress = http://localhost:5295
3 Send Request
4 GET {{ContosoPizza_HostAddress}}/weatherforecast/
5 Accept: application/json
6 ###
7
8 Send Request
9 GET {{ContosoPizza_HostAddress}}/pizza/
10 Accept: application/json
11 ###
12
```
- RESPONSE:** The response for the first request is shown:

```
1 HTTP/1.1 200 OK
2 Connection: close
3 Content-Type: application/json; charset=utf-8
4 Date: Tue, 24 Feb 2026 09:04:28 GMT
5 Server: Kestrel
6 Transfer-Encoding: chunked
7
8 [
9   {
10    "id": 1,
11    "name": "Classic Italian",
12    "isGlutenFree": false
13   },
14   {
15    "id": 2,
16    "name": "Veggie",
17    "isGlutenFree": true
18   }
19 ]
```
- TERMINAL:** The output shows the application building and starting successfully on port 5295.

```
ContosoPizza net8.0 succeeded (2.8s) -> bin\Debug\net8.0\ContosoPizza.dll
Building...
info: Microsoft.Hosting.Lifetime[14]
      Now listening on: http://localhost:5295
info: Microsoft.Hosting.Lifetime[0]
      Application started. Press Ctrl+C to shut down.
info: Microsoft.Hosting.Lifetime[0]
      Hosting environment: Development
info: Microsoft.Hosting.Lifetime[0]
      Content root path: C:\Users\Mayuresh\OneDrive\Desktop\ThinkBridge_January_2026\Week 6 Day 1\ContosoPizza
warn: Microsoft.AspNetCore.HttpsPolicy.HttpsRedirectionMiddleware[3]
      Failed to determine the https port for redirect.
```

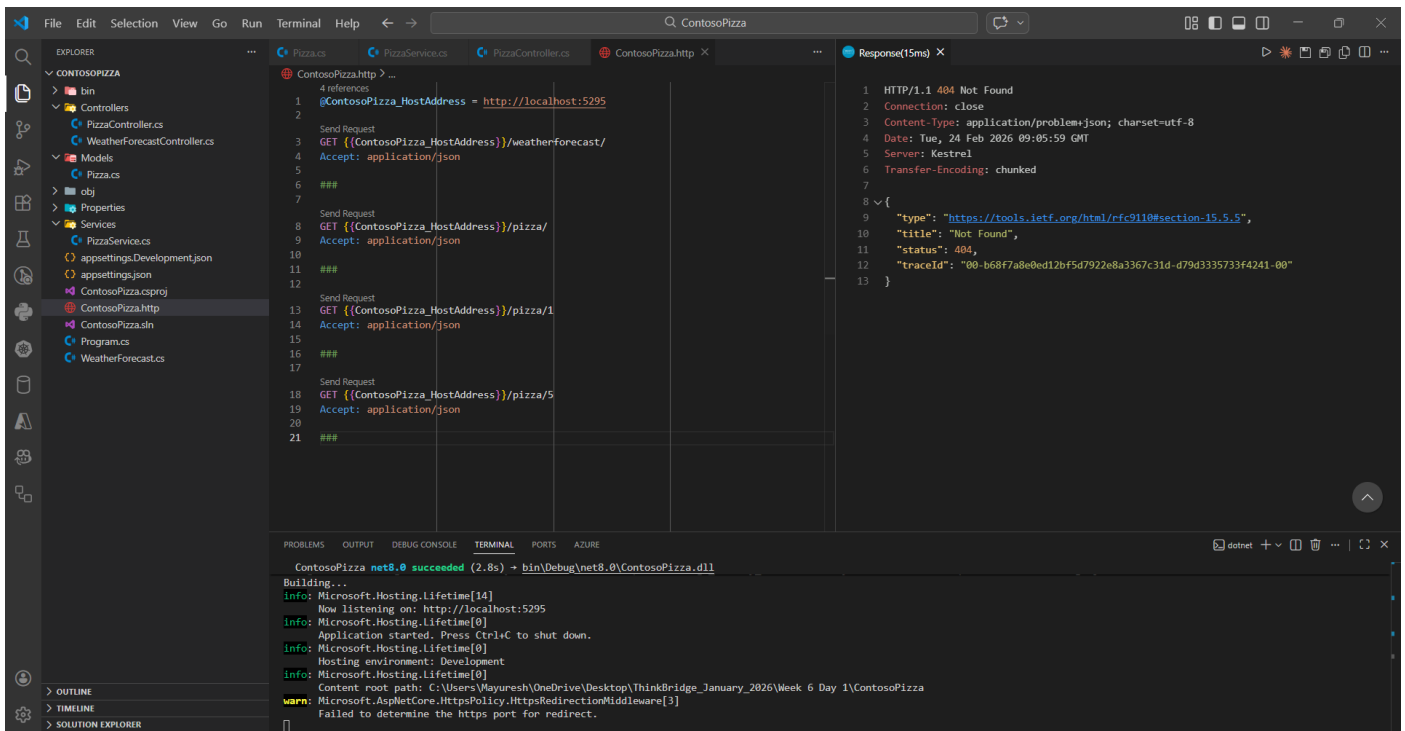
The screenshot shows the Visual Studio IDE with the following components:

- EXPLORER:** The project structure for 'CONTOSOPIZZA' is visible, including 'bin', 'Controllers' (with 'PizzaController.cs'), 'Models', 'obj', 'Properties', 'Services' (with 'PizzaService.cs'), 'appsettings.Development.json', 'appsettings.json', 'ContosoPizza.csproj', 'ContosoPizza.http', 'ContosoPizza.sln', 'Program.cs', and 'WeatherForecast.cs'.
- EDITOR:** The 'ContosoPizza.http' file is open, showing a REST client request:

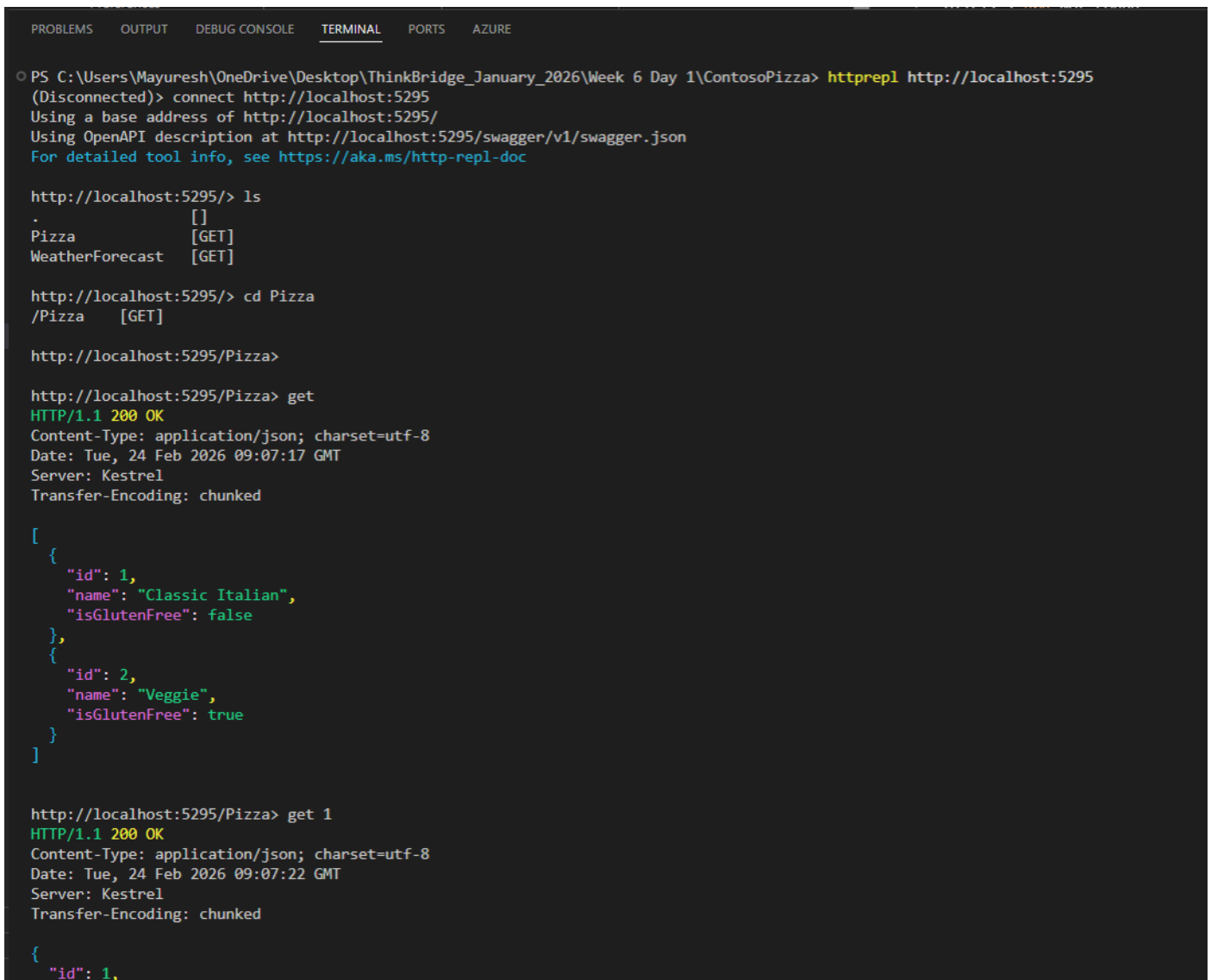
```
1 GET /pizza/
2 @ContosoPizza_HostAddress = http://localhost:5295
3 Send Request
4 GET {{ContosoPizza_HostAddress}}/weatherforecast/
5 Accept: application/json
6 ###
7
8 Send Request
9 GET {{ContosoPizza_HostAddress}}/pizza/
10 Accept: application/json
11 ###
12
13 Send Request
14 GET {{ContosoPizza_HostAddress}}/pizza/1
15 Accept: application/json
16 ###
```
- RESPONSE:** The response for the second request is shown:

```
1 HTTP/1.1 200 OK
2 Connection: close
3 Content-Type: application/json; charset=utf-8
4 Date: Tue, 24 Feb 2026 09:05:36 GMT
5 Server: Kestrel
6 Transfer-Encoding: chunked
7
8 [
9   {
10    "id": 1,
11    "name": "Classic Italian",
12    "isGlutenFree": false
13   }
14 ]
```
- TERMINAL:** The output shows the application building and starting successfully on port 5295.

```
ContosoPizza net8.0 succeeded (2.8s) -> bin\Debug\net8.0\ContosoPizza.dll
Building...
info: Microsoft.Hosting.Lifetime[14]
      Now listening on: http://localhost:5295
info: Microsoft.Hosting.Lifetime[0]
      Application started. Press Ctrl+C to shut down.
info: Microsoft.Hosting.Lifetime[0]
      Hosting environment: Development
info: Microsoft.Hosting.Lifetime[0]
      Content root path: C:\Users\Mayuresh\OneDrive\Desktop\ThinkBridge_January_2026\Week 6 Day 1\ContosoPizza
warn: Microsoft.AspNetCore.HttpsPolicy.HttpsRedirectionMiddleware[3]
      Failed to determine the https port for redirect.
```



Optional : Test the controller with Command Line HTTP Read-Eval-Print Loop (REPL)



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS AZURE

PS C:\Users\Mayuresh\OneDrive\Desktop\ThinkBridge_January_2026\Week 6 Day 1\ContosoPizza> httprepl http://localhost:5295

http://localhost:5295/Pizza> get 1
HTTP/1.1 200 OK
Content-Type: application/json; charset=utf-8
Date: Tue, 24 Feb 2026 09:07:22 GMT
Server: Kestrel
Transfer-Encoding: chunked

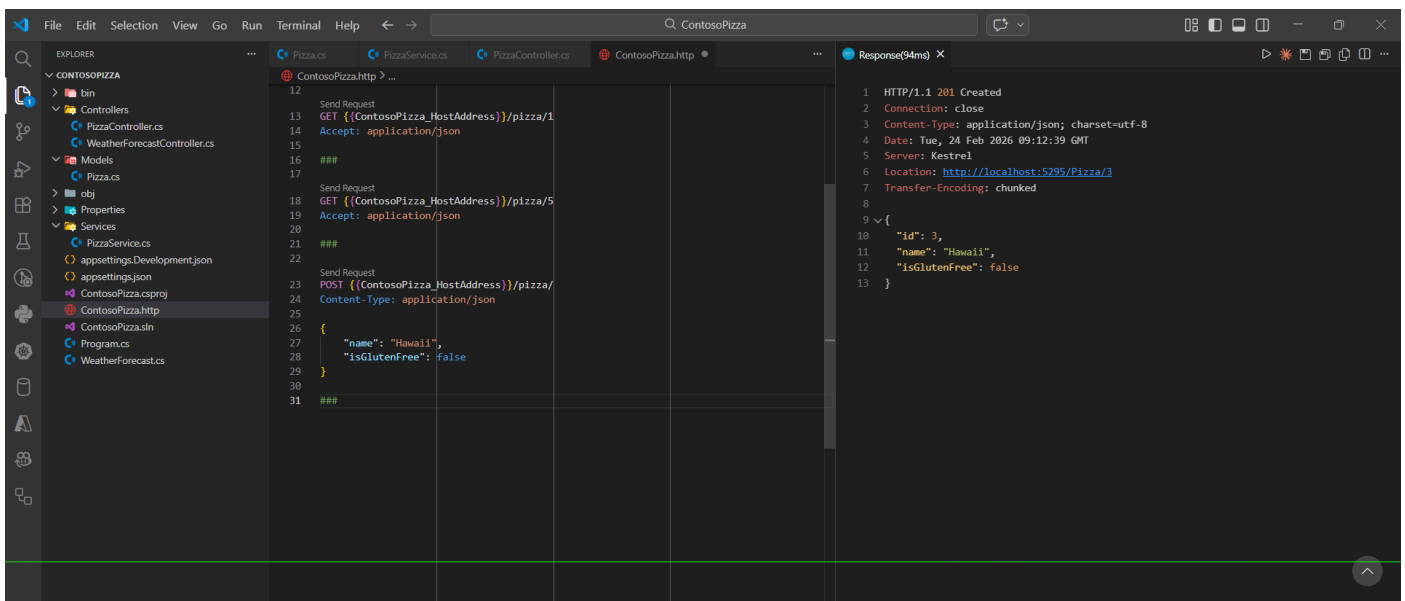
{
  "id": 1,
  "name": "Classic Italian",
  "isGlutenFree": false
}

http://localhost:5295/Pizza> get 5
HTTP/1.1 404 Not Found
Content-Type: application/problem+json; charset=utf-8
Date: Tue, 24 Feb 2026 09:07:28 GMT
Server: Kestrel
Transfer-Encoding: chunked

{
  "type": "https://tools.ietf.org/html/rfc9110#section-15.5.5",
  "title": "Not Found",
  "status": 404,
  "traceId": "00-00bb7887eb93e2889a27de61595f165e-9ae69f38d2909a4c-00"
}

http://localhost:5295/Pizza> 
```

Exercise - Implement CRUD operations



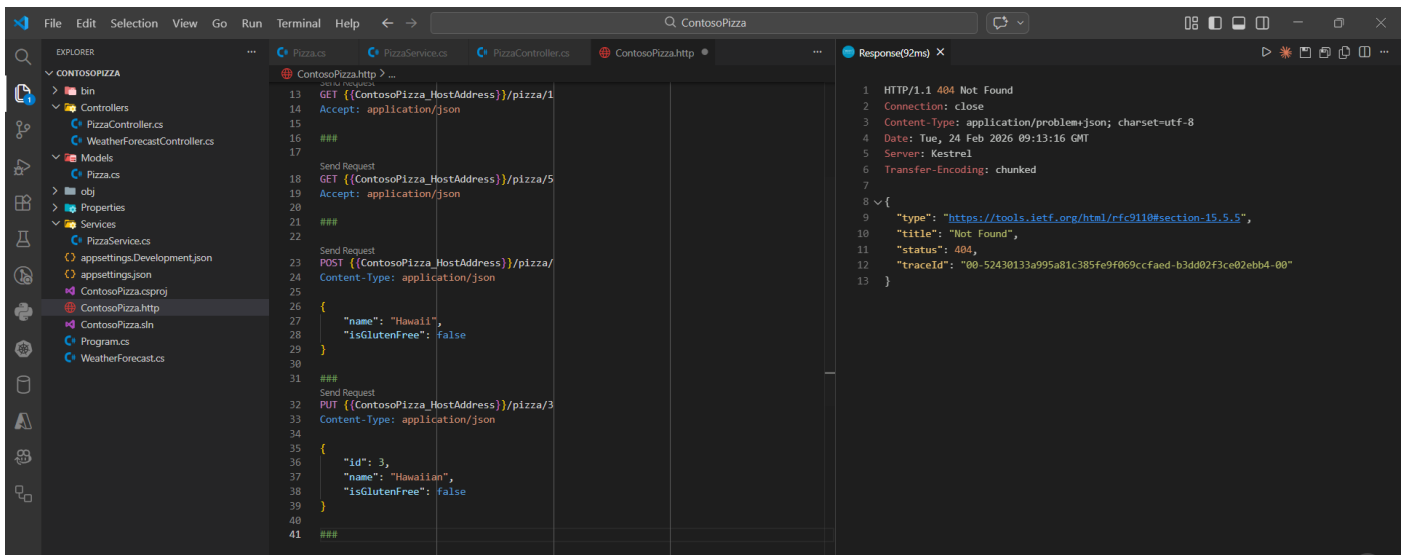
The screenshot shows the Visual Studio Code interface with the ContosoPizza project. The Explorer pane on the left displays the project structure, including folders for bin, Controllers, Models, and Services, and files for PizzaService.cs, appsettings.json, ContosoPizza.csproj, ContosoPizza.http, ContosoPizza.sln, Program.cs, and WeatherForecast.cs. The ContosoPizza.http file is open in the editor, showing a REST client configuration with a base URL of {{ContosoPizza_HostAddress}}/pizza/. The interface also shows a REST client response for a GET request to the /pizza/3 endpoint, returning a 201 Created status and a JSON object with id 3, name Hawaii, and isGlutenFree false.

```
File Edit Selection View Go Run Terminal Help < -> ContosoPizza

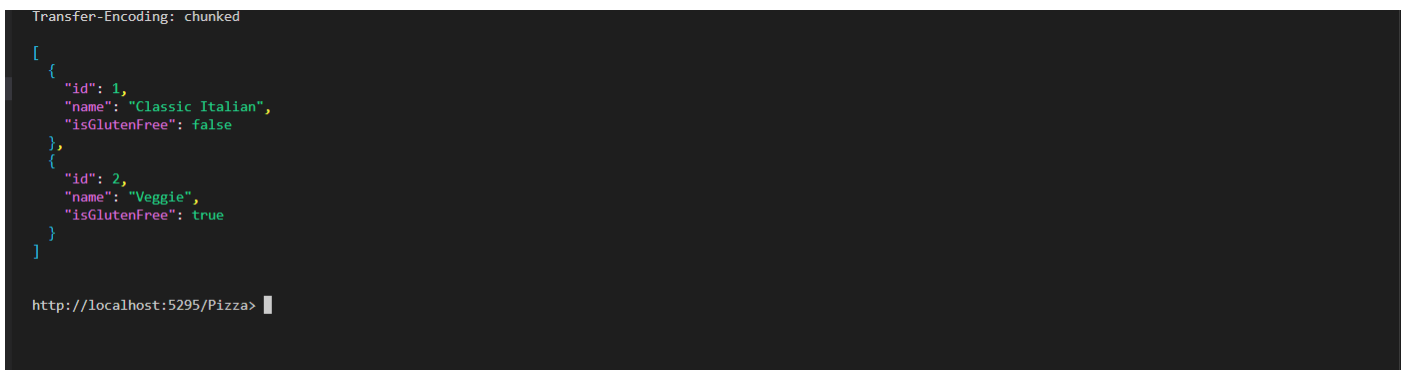
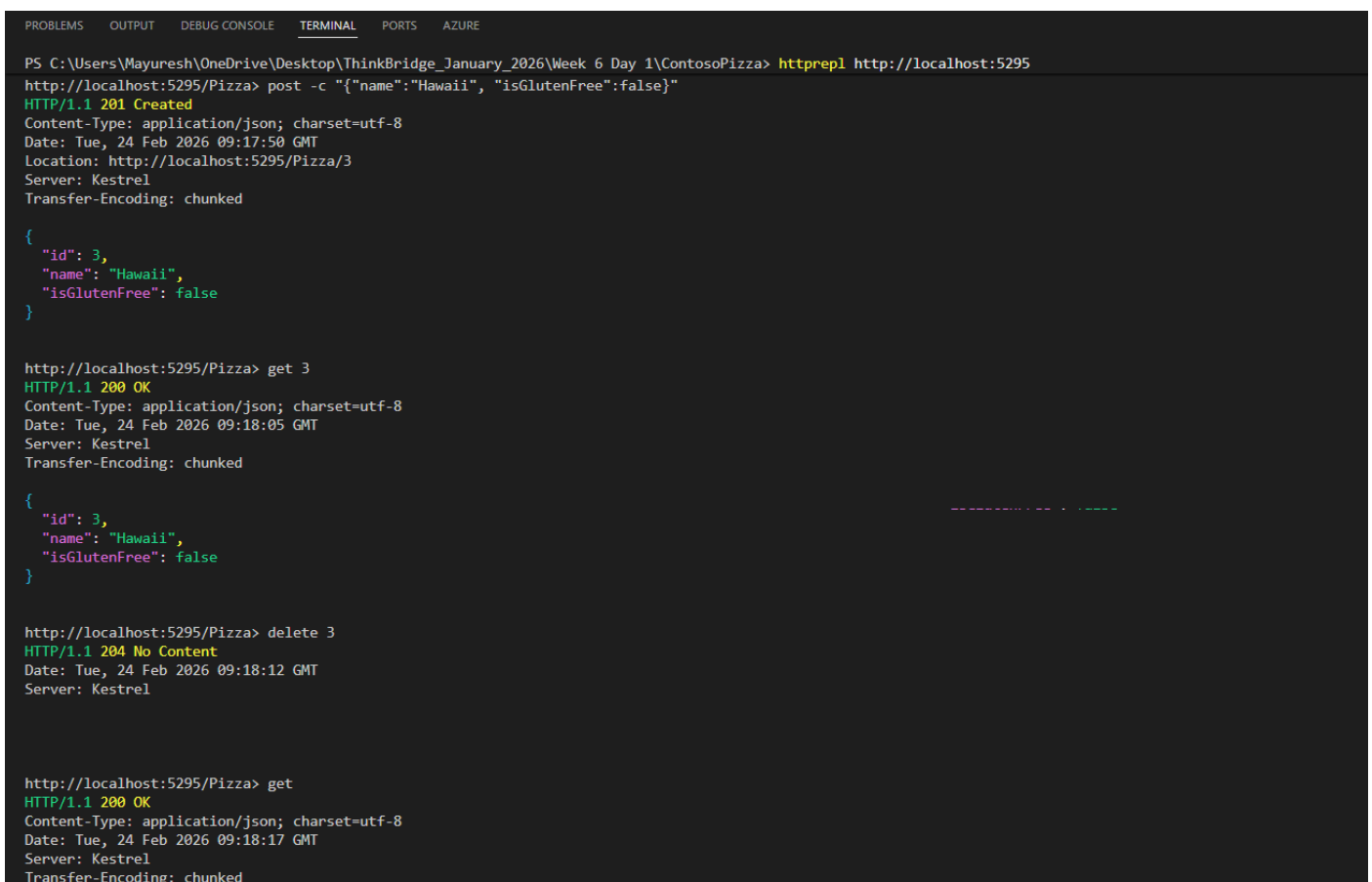
EXPLORER
  CONTOSOPIZZA
    bin
    Controllers
      PizzaController.cs
      WeatherForecastController.cs
    Models
      Pizza.cs
    obj
    Properties
    Services
      PizzaService.cs
    appsettings.Development.json
    appsettings.json
    ContosoPizza.csproj
    ContosoPizza.http
    ContosoPizza.sln
    Program.cs
    WeatherForecast.cs

ContosoPizza.http
12 ContosoPizza.http > ...
13 Send Request
14 GET {{ContosoPizza_HostAddress}}/pizza/1
15 Accept: application/json
16 ###
17
18 Send Request
19 GET {{ContosoPizza_HostAddress}}/pizza/5
20 Accept: application/json
21 ###
22
23 Send Request
24 POST {{ContosoPizza_HostAddress}}/pizza/
25 Content-Type: application/json
26 {
27   "name": "Hawaii",
28   "isGlutenFree": false
29 }
30
31 ###

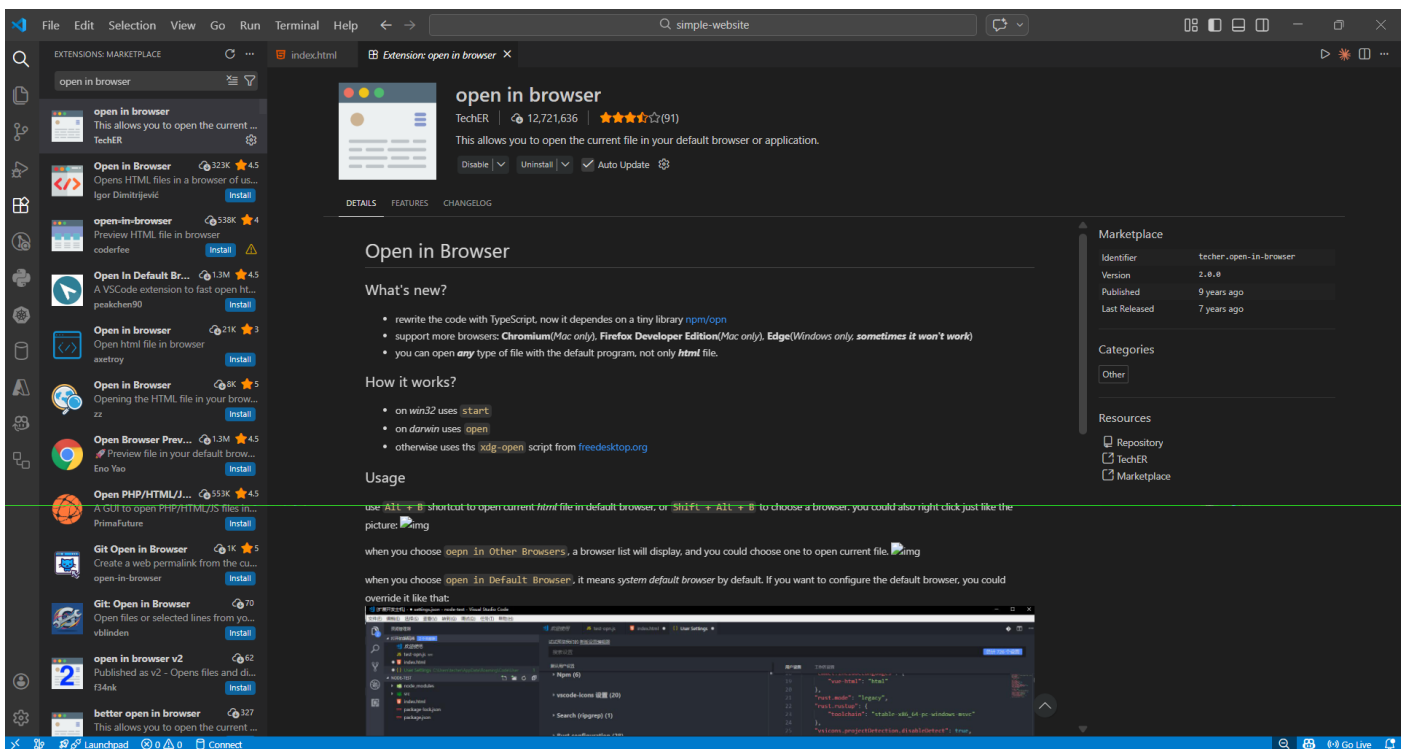
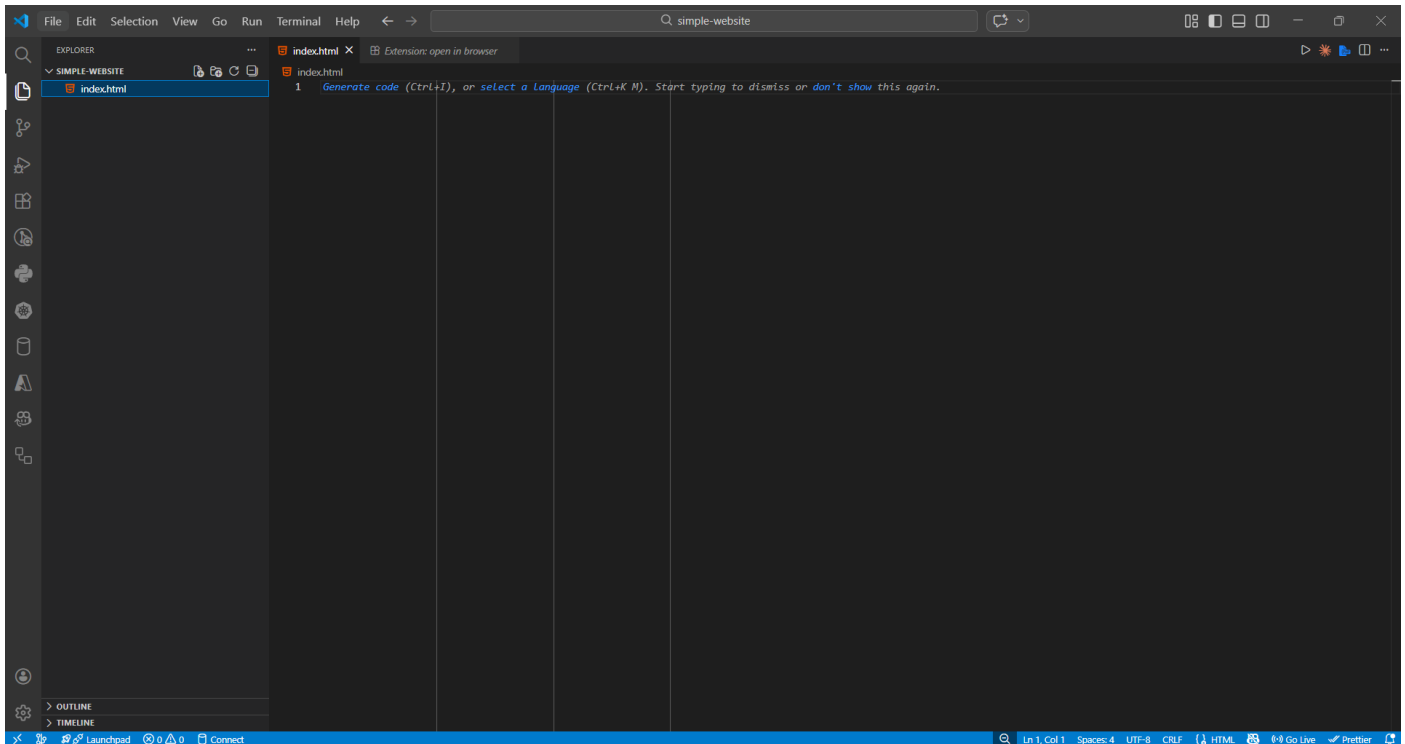
Response(94ms) X
1 HTTP/1.1 201 Created
2 Connection: close
3 Content-Type: application/json; charset=utf-8
4 Date: Tue, 24 Feb 2026 09:12:39 GMT
5 Server: Kestrel
6 Location: http://localhost:5295/Pizza/3
7 Transfer-Encoding: chunked
8
9 {
10   "id": 3,
11   "name": "Hawaii",
12   "isGlutenFree": false
13 }
```

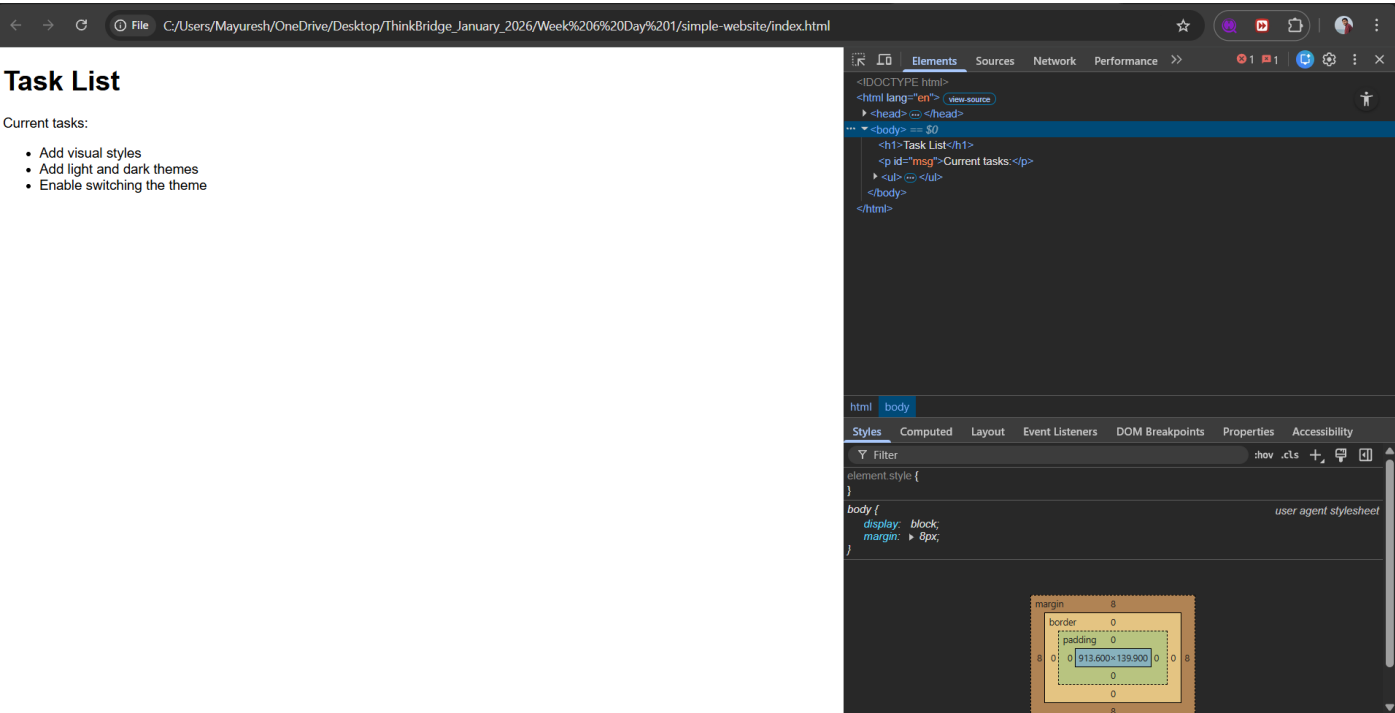
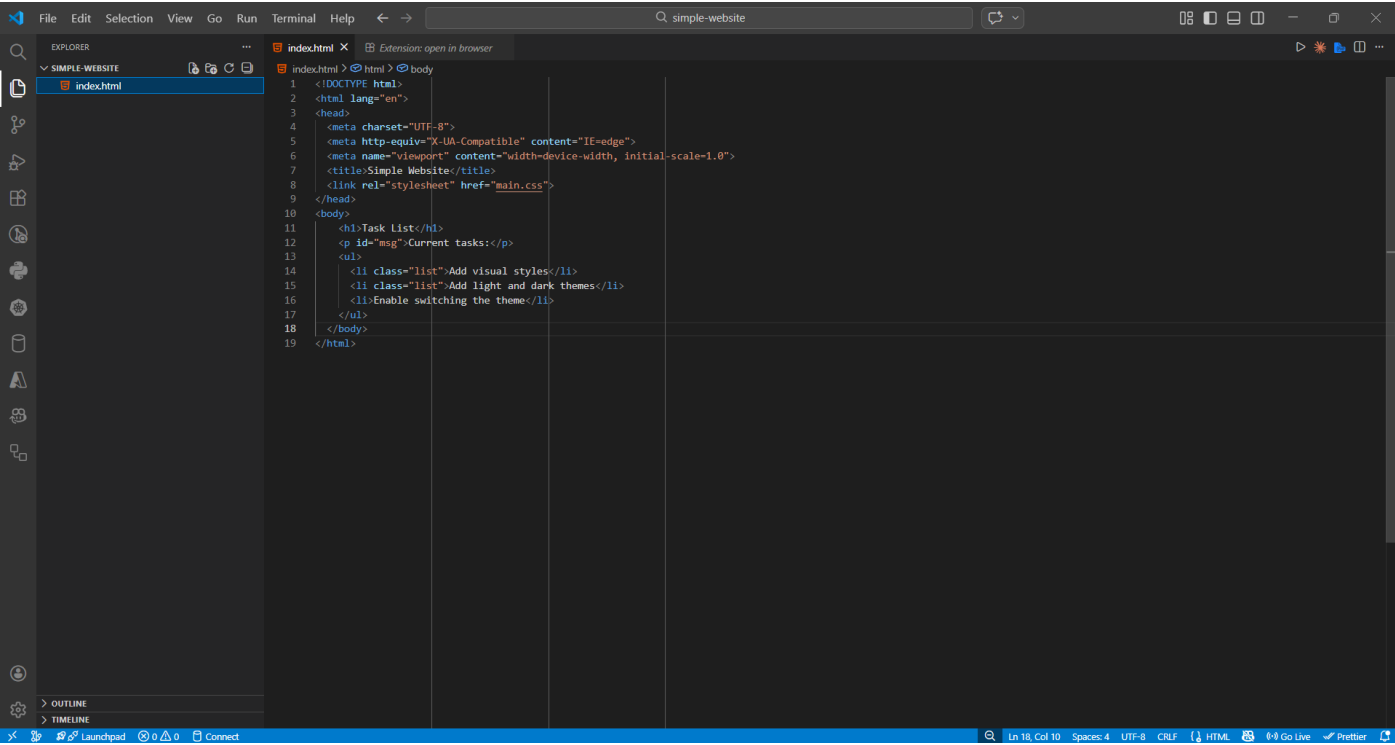
Optional: Test the finished web API with Command Line HTTPREPL



Exercise - Set up the structure of your web app



Exercise - Add basic HTML to your web app



Exercise - Style your HTML with CSS

Task List

Current tasks:

- Add visual styles
- Add light and dark themes
- Enable switching the theme

ElementsSourcesNetworkPerformanceConsole

<!DOCTYPE html>
<html lang="en">
<head>
</head>
<body class="light-theme">
<h1>Task List</h1>
<p id="msg">Current tasks:</p>

</body>
</html>

htmlbody.light-theme

StylesComputedLayoutEvent ListenersDOM BreakpointsPropertiesAccessibility

Filter

element style {
}
light-theme {
color: #000000;
background: #00ff00;
}
body {
font-family: monospace;
}
body {
display: block;
margin: 8px;
}

Task List

Current tasks:

- Add visual styles
- Add light and dark themes
- Enable switching the theme

ElementsSourcesNetworkPerformanceConsole

<!DOCTYPE html>
<html lang="en">
<head>
</head>
<body class="dark-theme">
<h1>Task List</h1>
<p id="msg">Current tasks:</p>

</body>
</html>

htmlbody.dark-theme

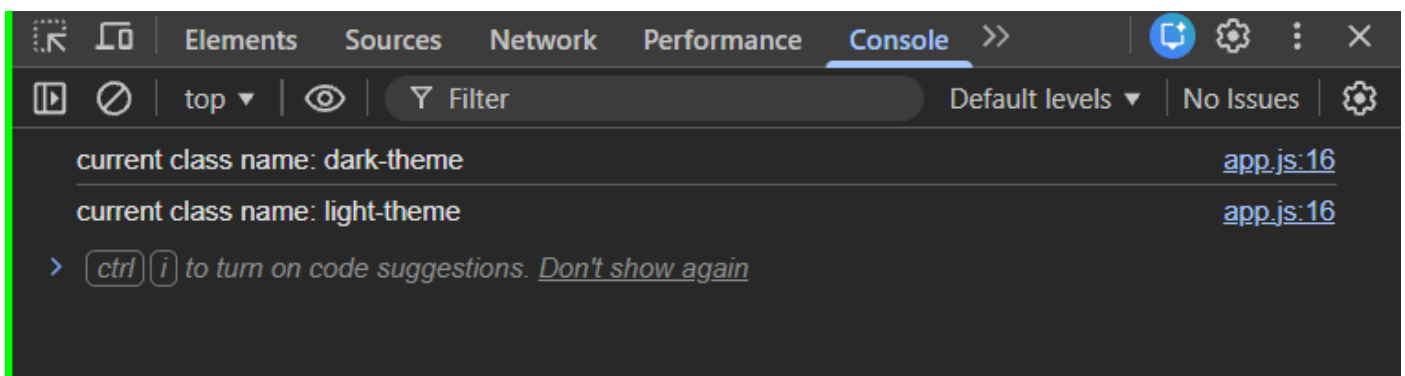
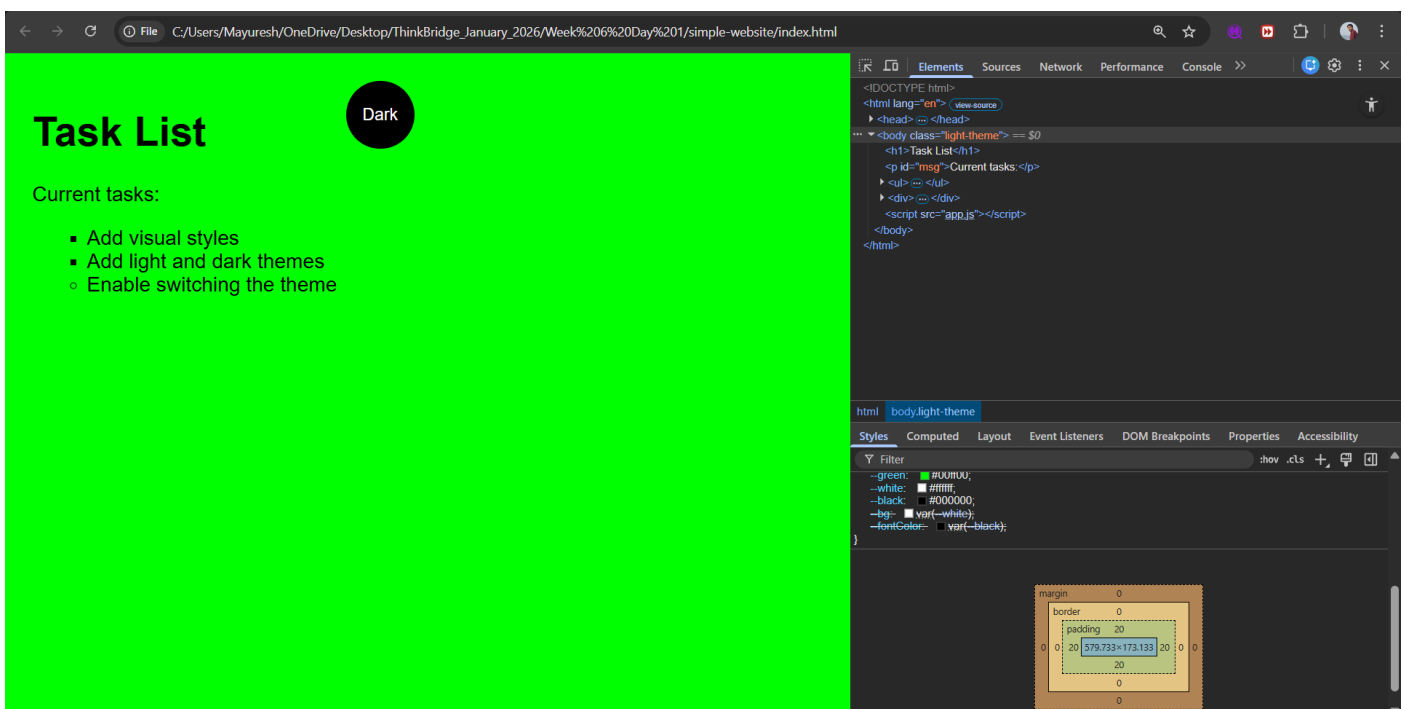
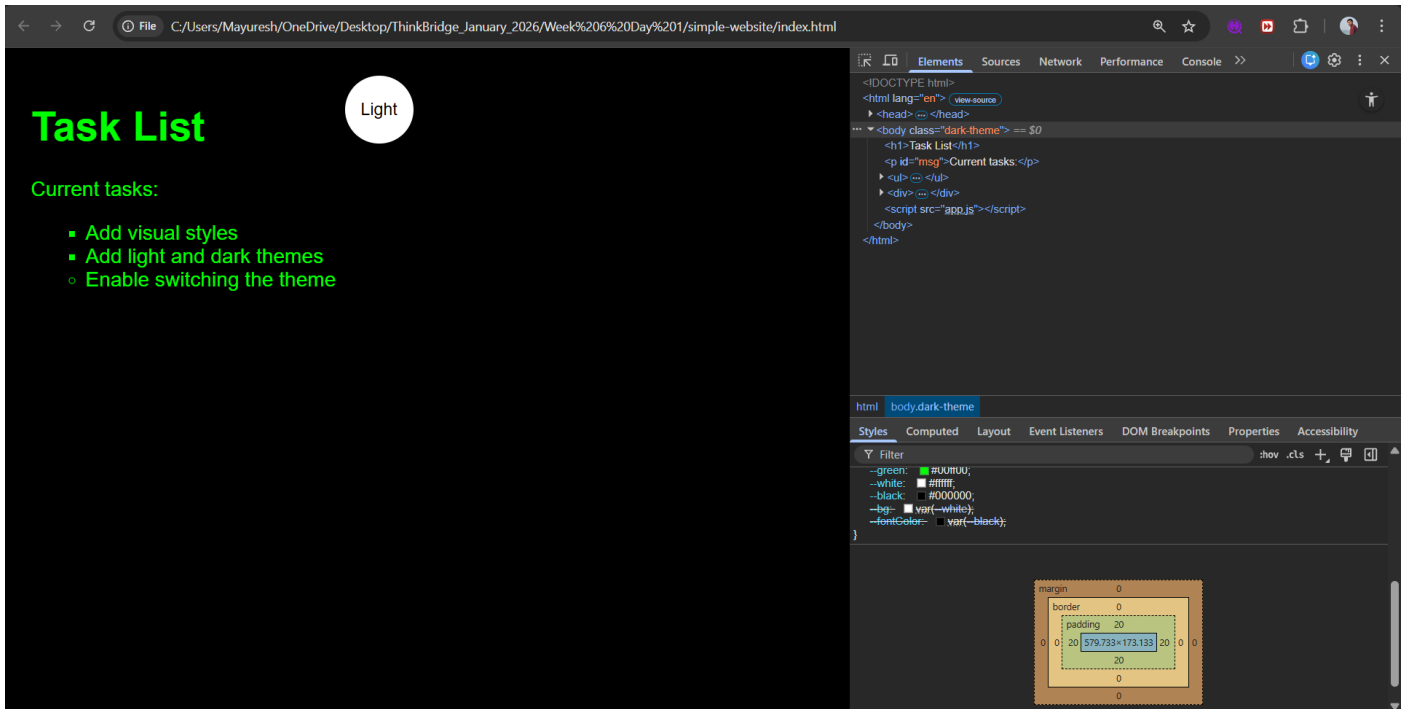
StylesComputedLayoutEvent ListenersDOM BreakpointsPropertiesAccessibility

Filter

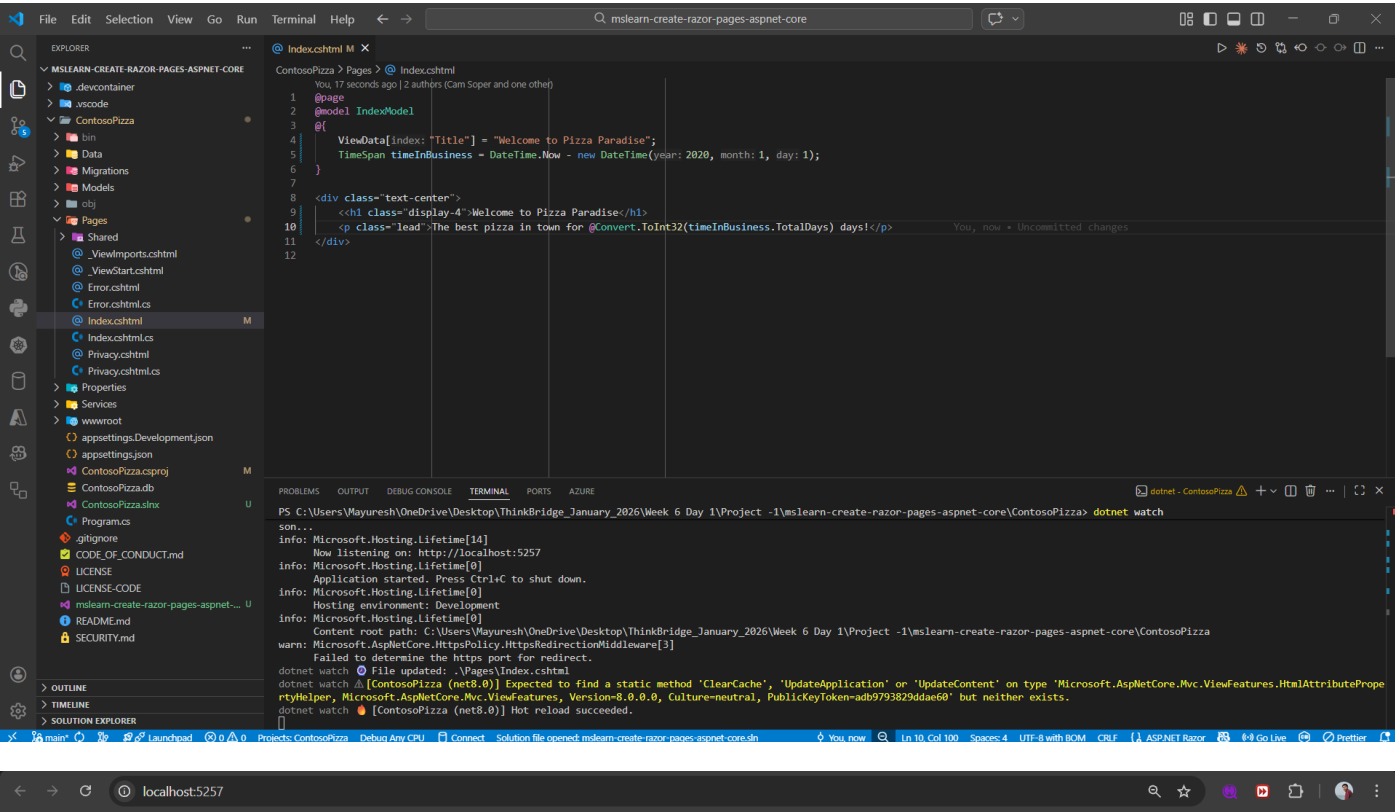
--black: #000000;
--bg: var(--white);
--fontColor: var(--black);
}

margin0
border0
padding20
0 0 20 20579.733px173.133px200
000

Exercise - Add interactivity with JavaScript



Exercise - Customize the project





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Items in this collection

SECTION 1

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Build .NET applications with C#

6 modules • 3 hr 14 min

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91% completed

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Write your first code using C# (Get started with C#, Part 1)

6 modules • 4 hr 37 min

10% completed

LEARNING PATH

Create and run simple C# console applications (Get started with C#, Part 2)